

MS 6015: Multivariate Statistical Methods for Business

Simple Linear Regression

March 2, 2026

Outline

Introduction

- Association between Variables

Simple Regression Model

- Regression for Prediction

- Capital Asset Pricing Model

- Inference in Regression

- Prediction Intervals

Curved Patterns

- Detecting Nonlinear Patterns

- Transformations

- Reciprocal Transformation

Regression Diagnostics

- Changing Variation

- Leveraged Outliers

- Dependent Errors and Time Series

References

Regression: General Introduction

- ▶ Regression analysis is the most widely used statistical tool for understanding relationships among variables
- ▶ It provides a conceptually simple method for investigating functional relationships between one or more factors and an outcome of interest
- ▶ The relationship is expressed in the form of an equation or a model connecting the response or dependent variable and one or more explanatory or predictor variable

Why Regression?

- ▶ What is the impact of MBA on income?
- ▶ How does the returns of a mutual fund relates to the market?
- ▶ Does Company A discriminate against women regarding salaries?
- ▶ Predict market price based on observed characteristics.

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Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

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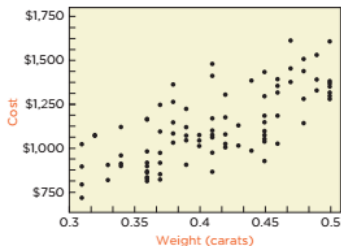
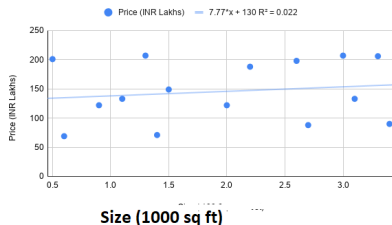
Leveraged Outliers

Dependent Errors and Time Series

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Visual test for association

- ▶ Do we see a pattern in the scatter plot?
 - ▶ In other words, if I know about the x -value, can I use it to say something about the y -value or guess y -value?



Describing association

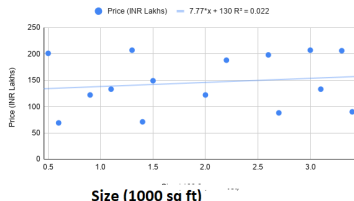
When describing association between variables in a scatter plot, there are four key questions¹ that need to be answered

1. **Direction:** Does the pattern trend up, down, or both?
2. **Curvature:** Does the pattern appear to be linear or does it curve?
3. **Variation:** Are the points tightly clustered along the pattern?
4. **Outliers:** Did you find something unexpected?

¹Stine, Robert, and Dean Foster. Statistics for Business: Decision Making and. Addison-Wesley, 2011.

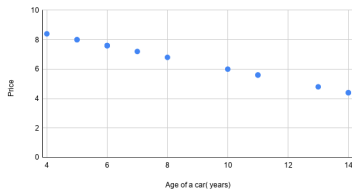
Describing association: Direction

Does the pattern trend up, down, or both?



Up

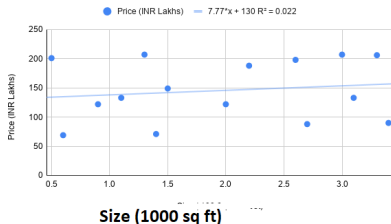
Price vs. Age of a car(years)



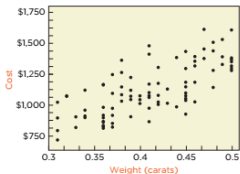
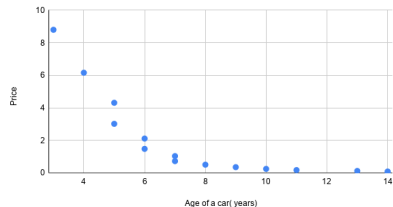
Down

Describing association: Curvature

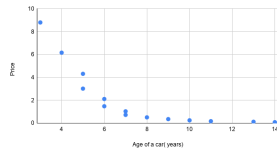
Does the pattern appear to be linear or does it curve?



Price vs. Age of a car (years)

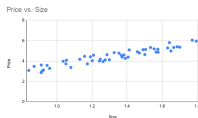


Price vs. Age of a car (years)

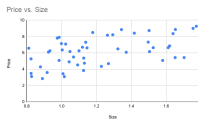


Describing association: Variation

Are the points tightly clustered along the pattern?



Tightly clustered

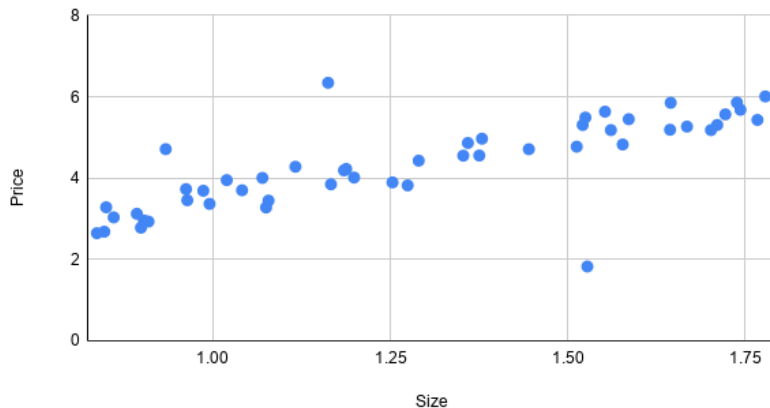


Variable

Describing association: Outliers

Did you find something unexpected?

Price vs. Size



Measures of association

How do we measure the strength of association between two variables?

1. Covariance
2. Correlation

Example: Predicting House Prices

Problem:

- ▶ Predict market price based on observed characteristics

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Solution:

- ▶ Look at property sales data where we know the price and some observed characteristics.
- ▶ Build a decision rule that predicts price as a function of the observed characteristics.

What characteristics do we use?

What are the variables of interest

- ▶ Many factors or variables affect the price of a house : Size, number of baths, garage, neighbourhood etc
- ▶ Develop a specific quantitative measure of these variables: Easy to quantify price and size but what about other variables such as aesthetics, workmanship, etc?

Keep things super simple, let's focus only on size.

- ▶ The value that we seek to predict is called the dependent (or output) variable, and we denote this:

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$$Y = \text{price of house (units of currency)}$$

- ▶ The variable that we use to guide prediction is the explanatory (or input) variable, and this is labelled as

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$$X = \text{size of house (e.g. thousands of square feet)}$$

The Simple Regression Model

- ▶ Simple Regression Model (SRM): model for the association in the population between an explanatory variable x and response y .
- ▶ Consider the data to be a sample from a population.

The Simple Linear Regression Model

- ▶ The equation of the SRM describes how the conditional mean of Y depends on X .

$$Y = \beta_0 + \beta_1 X + \epsilon$$

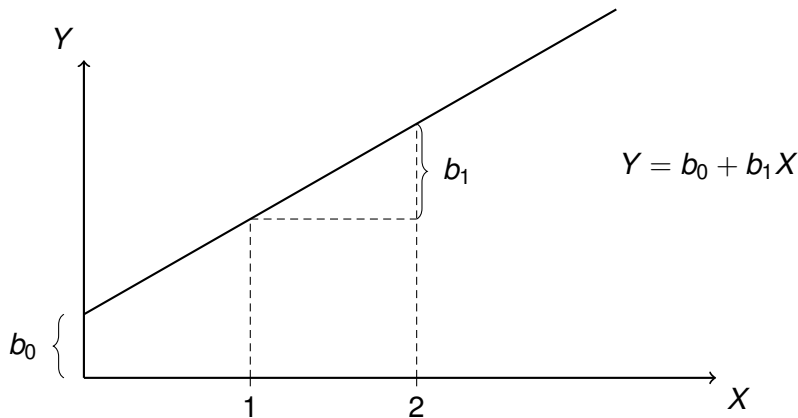
- ▶ The SRM shows that these means lie on a line with intercept β_0 and slope β_1 :

$$\mu_{Y|X} = E(Y|X = x) = \hat{\beta}_0 + \hat{\beta}_1 x = b_0 + b_1 x$$

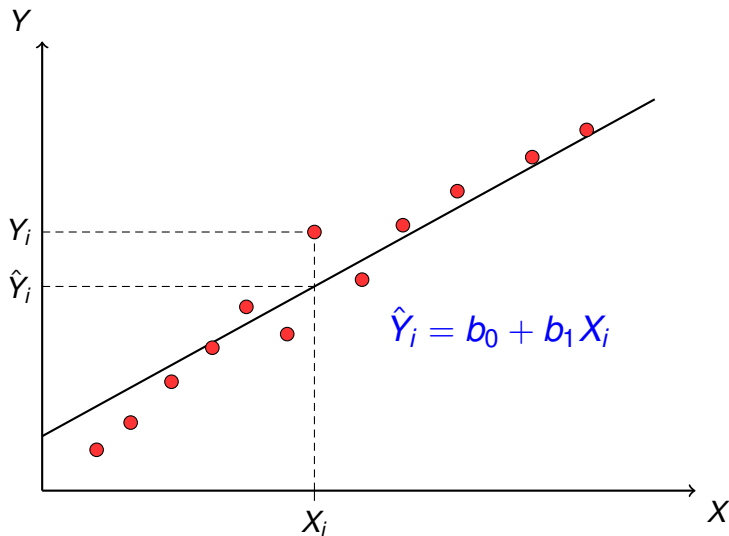
Least Squares Method to Find the Regression Line

- ▶ Need to estimate the slope and intercept.
- ▶ Fit a line in which minimizes the amount by which the actual value differs from the fitted value
- ▶ The difference is called the Residual

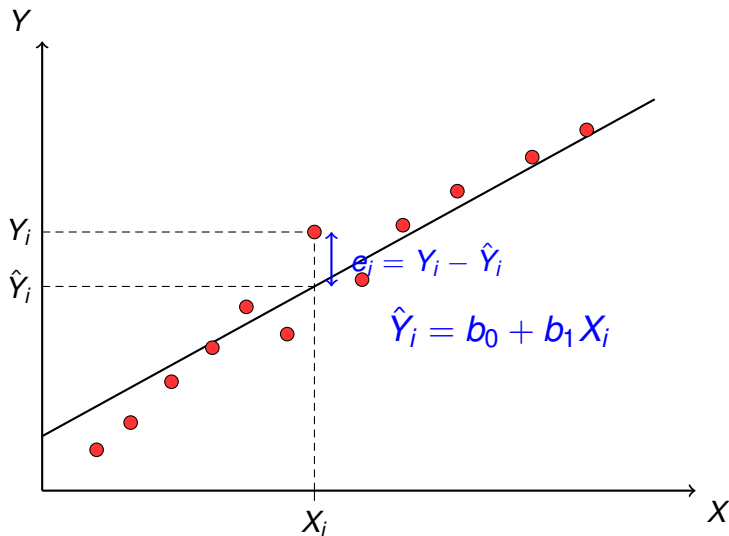
Linear model: $Y = b_0 + b_1 X$



Fitted value illustration



Fitted value and residual



Method of Least Squares

Minimize the size of all residuals

- ▶ If all residuals were zero-Perfect fit!!
- ▶ Give weights to residuals
- ▶ Minimize total residuals to get best fit
- ▶ Least squares method chooses b_0 and b_1 to minimize

$$\begin{aligned}\sum_{i=1}^n e_i^2 &= e_1^2 + e_2^2 + \dots + e_n^2 \\ &= (Y_1 - \hat{Y}_1)^2 + (Y_2 - \hat{Y}_2)^2 + \dots + (Y_N - \hat{Y}_n)^2\end{aligned}$$

Least Squares Estimates

The formula for b_0 and b_1 that minimize the least squares criterion are

$$b_1 = r_{XY} \frac{s_Y}{s_X}; \quad b_0 = \bar{Y} - b_1 \bar{X}$$

where

- ▶ $\bar{X}$ and $\bar{Y}$ are the sample mean of X and Y
- ▶ $\text{corr}(X, Y) = r_{XY}$ is sample correlation
- ▶ s_X and s_Y are sample standard deviation of X and Y

Residuals and X

$$Y = \hat{Y} + \text{residual}$$

where

$$\text{corr}(\hat{Y}, X) = 1$$

Error is unrelated to X ; $\text{corr}(X, e) = 0$.

Decomposing the Variance

How well does the least square line explain variation in Y

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How well does the least square line explain variation in Y

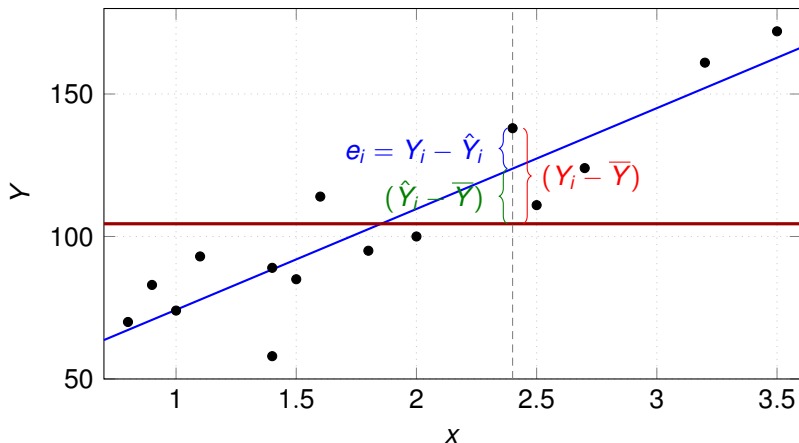
$$\begin{aligned} \text{Var}(Y) &= \text{Var}(\hat{Y} + e) \\ &= \text{Var}(\hat{Y}) + \text{Var}(e) + \text{cov}(\hat{Y}, e) \\ &= \text{Var}(\hat{Y}) + \text{Var}(e); \text{ as } \text{corr}(\hat{Y}, e) = 0 \end{aligned}$$

Decomposing the Variance

How well does the least square line explain variation in Y

$$\begin{aligned} \text{Var}(Y) &= \text{Var}(\hat{Y} + e) \\ &= \text{Var}(\hat{Y}) + \text{Var}(e) + \text{cov}(\hat{Y}, e) \\ &= \text{Var}(\hat{Y}) + \text{Var}(e); \text{ as } \text{corr}(\hat{Y}, e) = 0 \\ \sum_{i=1}^n (Y_i - \bar{Y})^2 &= \sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2 + \sum_{i=1}^n (e_i - \bar{e})^2 \\ &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (e_i)^2 \end{aligned}$$

Decomposition: $(Y_i - \bar{Y}) = (\hat{Y}_i - \bar{Y}) + e_i$



Decomposing the Variance

$$\underbrace{\sum_{i=1}^n (Y_i - \bar{Y})^2}_{\text{Total Sum of Squares}} = \underbrace{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}_{\text{Regression SS}} + \underbrace{\sum_{i=1}^n (e_i)^2}_{\text{Error SS}}$$

$$SST = SSR + SSE$$

- ▶ *SSR* : Variation in Y explained by the regression line.
- ▶ *SSE* : Variation in Y that is left unexplained.

Decomposing the Variance

$$\underbrace{\sum_{i=1}^n (Y_i - \bar{Y})^2}_{\text{Total Sum of Squares}} = \underbrace{\sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2}_{\text{Regression SS}} + \underbrace{\sum_{i=1}^n (e_i)^2}_{\text{Error SS}}$$

$$SST = SSR + SSE$$

- ▶ SSR : Variation in Y explained by the regression line.
- ▶ SSE : Variation in Y that is left unexplained.
- ▶ If $SSR = SST \implies$ **Perfect Fit!!**

Measuring Goodness of Fit: Coefficient of Determination

The coefficient of determination, denoted by R^2 measures goodness of fit:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

- ▶ $0 < R^2 < 1$
- ▶ The closer R^2 is to 1, the better the fit.
- ▶ $R^2 = r_{XY}^2$

- └ Simple Regression Model
 - └ Regression for Prediction

Outline

Introduction

Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

Regression Diagnostics

Changing Variation

Leveraged Outliers

Dependent Errors and Time Series

References

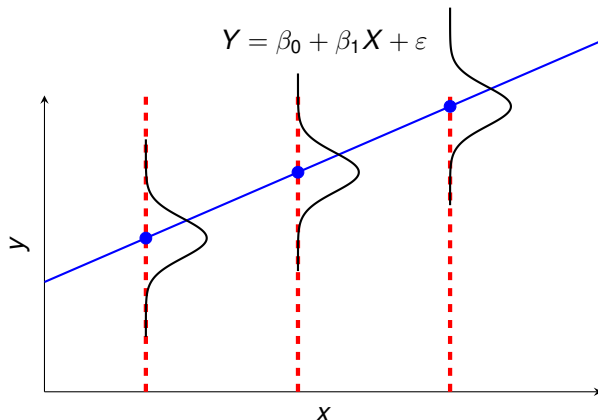
Prediction

A prediction rule is any function where you input X and it outputs $\hat{Y}$ as a predicted response at X .

Prediction Intervals

- ▶ What we want to know
 - ▶ What value of Y can we expect for a given X ?
 - ▶ How sure are we about this forecast?
- ▶ In other words, we want to measure accuracy of forecast
 - ▶ How much uncertainty is there in the forecast
- ▶ Prediction Interval: Probable range for Y values given X

Conditional Distribution



The conditional distribution of Y given X is Normal

$$Y|X \sim N(\beta_0 + \beta_1 X, \sigma^2)$$

Why $\varepsilon \sim N(0, \sigma^2)$?

- ▶ $E(\varepsilon) = 0 \Leftrightarrow E(Y|X) = \beta_0 + \beta_1 X$
- ▶ $E(Y|X)$ is conditional expectation of Y given X
- ▶ By central limit theorem, can assume Normality.
- ▶ Given that I know X what kind of Y can I expect?
- ▶ Regression models are really all about modeling the conditional distribution of Y given X

Assumptions

The SRM makes three assumptions about ϵ

1. **Independent**. Errors are independent of each other.
2. **Equal variance**. All errors have the same variance,
 $Var(\epsilon) = \sigma_{\epsilon}^2$.
3. **Normal**. The errors are normally distributed.

Observations

Two general points

- ▶ If X has no forecasting power, then the marginal and conditionals will be the same.
- ▶ If X has some forecasting information,
 - ▶ then conditional means will be different than the marginal or overall mean and
 - ▶ conditional standard deviation of Y given X will be less than the marginal standard deviation of Y .

Conditions for the SRM - Checklist

- ▶ Is the association between y and x linear?
- ▶ Have lurking variables been ruled out?
- ▶ Are the errors evidently independent?
- ▶ Are the residuals nearly normal?
- ▶ Are the variances of the residuals similar?

└ Simple Regression Model

└ Capital Asset Pricing Model

Outline

Introduction

Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

Regression Diagnostics

Changing Variation

Leveraged Outliers

Dependent Errors and Time Series

References

Capital Asset Pricing Model

- ▶ The CAPM describes the relationship between returns on a speculative asset and returns on the whole stock market.
- ▶ Underlying theory suggests that market rewards investors for taking market risks.
- ▶ Can formulate the CAPM as a Simple Regression

$$R_t = \beta_0 + \beta_1 M_t + \epsilon_t$$

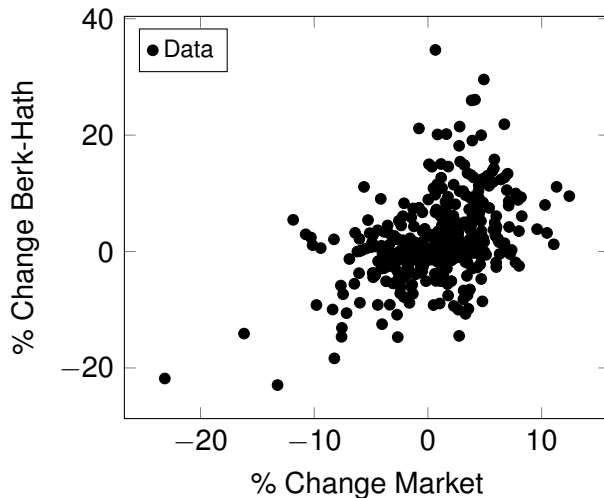
- ▶ R_t : Percentage change in value of asset- return on stock.
- ▶ M_t : Percentage change in value of the whole stock market

Example: Berkshire Hathaway Stock

How can we test the CAPM (Capital Asset Pricing Model) for Berkshire Hathaway stock?

- ▶ Formulate the simple regression with percentage change in Berkshire Hathaway stock as y and the percentage change in value of the whole stock market as x
- ▶ Use inference related to regression: standard errors, confidence intervals and hypothesis tests

CAPM Scatter Plot



CAPM Contd

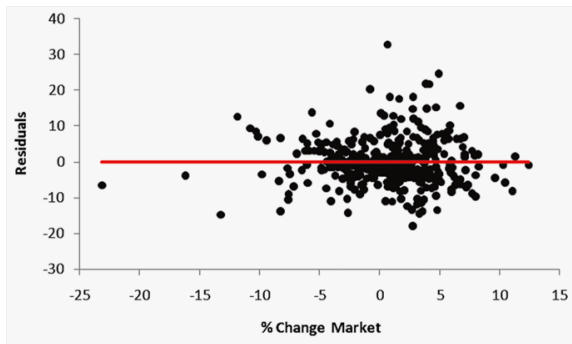
- ▶ According to CAPM, the mean of ϵ_t is zero, and

$$\beta_0 = 0$$

- ▶ On average, if the return on the market is zero, then we expect the return on a stock to be zero as well.
- ▶ Use Regression Analysis to test this theory

- └ Simple Regression Model
- └ Capital Asset Pricing Model

Conditions for the SRM- CAPM-Residual Plot

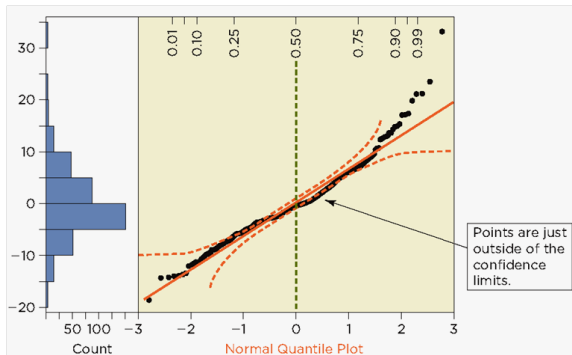


Linearity condition is satisfied; no pattern in the residuals. Data are shifted to the right because of two outliers (well-known declines in the market).

Conditions for the SRM- CAPM

- ▶ No obvious lurking variable (according to CAPM theory).
- ▶ Similar variances condition is satisfied. Check the plot of residuals versus x for any fan shaped pattern (none visible).

Conditions for the SRM- CAPM-Normality



The residuals are not normally distributed. Check sample size condition (satisfied) to use CLT: The sample size should be larger than 10 times the absolute value of the kurtosis K_4 of the residuals.

- ▶ Plot y versus x and verify the association appears linear.
- ▶ If pattern is linear, fit the least squares regression line and obtain residuals
- ▶ Plot residuals against explanatory variable- plot should have no pattern. Note presence of outliers as well.
- ▶ If data measured over time, check for timeplot for dependence
- ▶ Inspect histogram and quantile plot of residuals to check normality.

Assumptions-Summary

Assumption	Diagnostic	Problem
Linearity	Scatter Plots	Nonlinear
Independence	Plots showing trend	Autocorrelation
Constant Variance	Residual Plots	Heteroscedasticity
Normality	Quantile Plots	Outliers, Skewness

Outline

Introduction

Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

Regression Diagnostics

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Leveraged Outliers

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References

Parameters and Estimates for SRM

Data		SRM
b_0	Intercept	β_0
b_1	Slope	β_1
$\hat{y}$	Line	$\mu_{y x}$
e	Deviation	ε
s_e	$SD(\varepsilon)$	σ_ε

Standard Errors

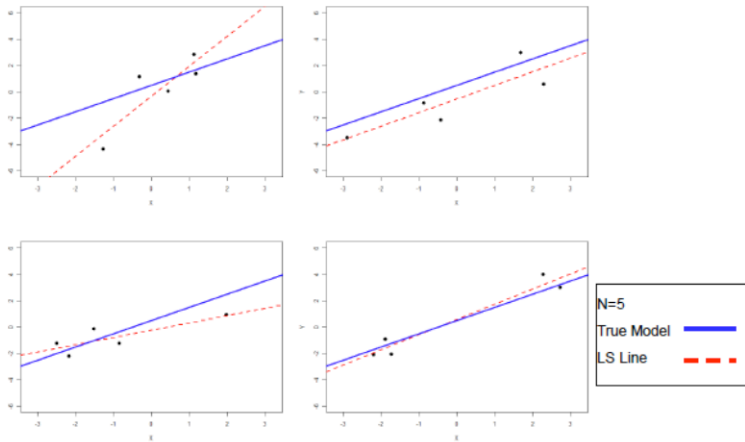
- ▶ Describe the sample-to-sample variability of b_0 and b_1
- ▶ The estimated standard error of b_1 is

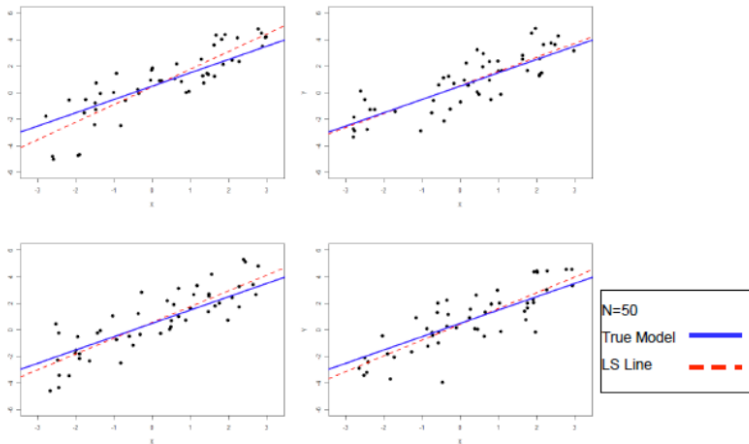
$$se(b_1) = \frac{s_e}{\sqrt{n-1}} \times \frac{1}{s_x} \approx \frac{s_e}{\sqrt{n}} \times \frac{1}{s_x}$$

Estimated Standard Error of b_1

Influenced by:

- ▶ Standard deviation of the residuals. As it increases, the standard error increases.
- ▶ Sample size. As it increases, the standard error decreases.
- ▶ Standard deviation of x . As it increases, the standard error decreases.





Software Results for CAPM Example

r^2	0.187872
s_e	6.517409
n	375

Term		Estimate	Std Error	<i>t</i> -Statistic	<i>p</i> -Value
Intercept	b_0	1.3962046	0.339682	4.11	<.0001
Market	b_1	0.7223495	0.077763	9.29	<.0001

Confidence Intervals: CAPM Example

- ▶ The 95% confidence interval for β_1 is

$$0.7223495 \pm 1.97 \times 0.07763 \approx [0.569, 0.876]$$

- ▶ The 95% confidence interval for β_0 is

$$1.3962046 \pm 1.97 \times 0.339682 \approx [0.727, 2.065]$$

Hypothesis Tests

- ▶ To test $H_0 : \beta_1 = 0$ Use $t = \frac{b_1}{se(b_1)}$
- ▶ To test $H_0 : \beta_0 = 0$ Use $t = \frac{b_0}{se(b_0)}$

Hypothesis Tests : CAPM Example

- ▶ The t -statistic of 9.29 with p -value of < 0.0001 indicates that the slope is significantly different from zero.
- ▶ The t -statistic of 4.11 with p -value of < 0.0001 indicates that the intercept is significantly different from zero

Interpretations for the CAPM Model-Beta

- ▶ The null hypothesis $H_1 : \beta_1 = 0$ If this were true then there is no linear relationship between X and Y .
- ▶ For the example this would imply that returns on Berkshire Hathaway are unrelated to returns on the market.
- ▶ The t -statistic tells us that the beta for the Berkshire Hathaway stock is not zero

Equivalent Inferences for the SRM

Reject the claim that a parameter in SRM equals zero with a 95% confidence(a 5% chance of Type I error) if

- ▶ Zero lies outside the 95% confidence interval for the parameter
- ▶ The absolute value of the associated t -statistic is larger than 2
- ▶ The p -value reported with the t -statistic is less than 0.05

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Introduction

Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

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References

Prediction Intervals

- ▶ Regression is used to predict the response for new, unobserved cases.
- ▶ SRM provides a framework that predicts y and anticipates the accuracy of prediction.
- ▶ SRM Model

$$y_{new} = \beta_0 + \beta_1 x_{new} + \epsilon_{new}; \epsilon_{new} \sim N(0, \sigma_\epsilon^2)$$

Prediction Interval vs Confidence Interval

- ▶ Prediction interval: an interval designed to hold a fraction (usually 95%) of the values of the response for a given value of x .
- ▶ A prediction interval differs from a confidence interval because it makes a statement about the location of a new observation rather than a parameter of a population.

Prediction Interval

- The 95% prediction interval for y_{new} is

$$\hat{y}_{new} \pm t_{0.025, n-2} se(\hat{y}_{new})$$

where $\hat{y}_{new} = b_0 + b_1 x_{new}$ and

$$se(\hat{y}_{new}) = s_e \sqrt{1 + \frac{1}{n} + \frac{(x_{new} - \bar{x})^2}{(n-1)s_x^2}}$$

- ▶ A simple approximation for a 95% prediction interval is .

$$\hat{y} \pm 2s_e$$

- ▶ Prediction intervals are reliable within the range of observed data. They are also sensitive to the assumptions of constant variance and normality.

Best Practices

- ▶ Verify that your model makes sense, both visually and substantively.
- ▶ Consider other possible explanatory variables.
- ▶ Check the conditions, in the listed order.
- ▶ Use confidence intervals to express what you know about the slope and intercept.
item Check the assumptions of the SRM carefully before using prediction intervals.

Pitfalls

- ▶ Don't overreact to residual plots.
- ▶ Do not mistake varying amounts of data for unequal variances.
- ▶ Do not confuse confidence intervals with prediction intervals.
- ▶ Do not expect that R^2 and s_e must improve with a larger sample.

Example-Cars

- ▶ One way to improve mileage is to reduce weight of the car
- ▶ Lighter materials cost more than heavier materials of comparable strength- Aluminium and composites are more expensive than steel.
- ▶ Manufacturers want to see evidence of benefit.

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Transformations

Reciprocal Transformation

Regression Diagnostics

Changing Variation

Leveraged Outliers

Dependent Errors and Time Series

References

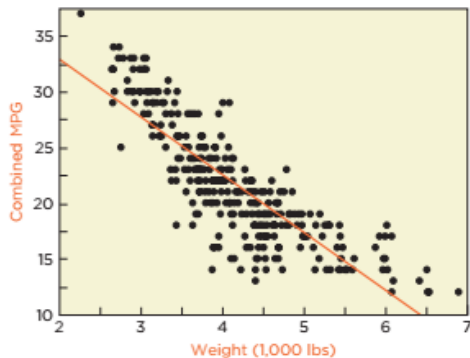
Detecting Nonlinear Patterns

- ▶ What improvement in mileage should a manufacturer expect from reducing the weight of a car?
- ▶ Use regression analysis to find an equation that summarizes the association between gas mileage and weight
- ▶ Is the pattern of association linear or curved?

Recognizing Nonlinearity

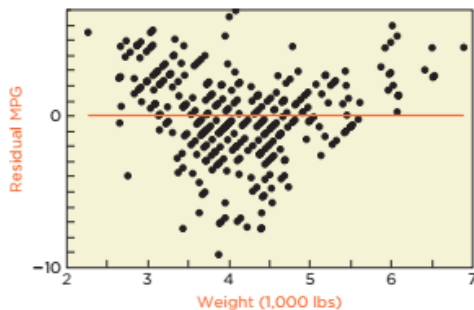
- ▶ Will changes in the explanatory variable result in equal sized changes in the estimated response, regardless of the value of X ?
- ▶ Does trimming 200 pounds from a large SUV have the same effect on mileage as trimming 200 pounds from a small compact?

Scatterplot with Fitted Line



- └ Curved Patterns
- └ Detecting Nonlinear Patterns

Residual Plot



Easier to spot curved pattern in the residuals. The association in the data is not straight enough to be summarized with a line.

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Introduction

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Simple Regression Model

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Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

Regression Diagnostics

Changing Variation

Leveraged Outliers

Dependent Errors and Time Series

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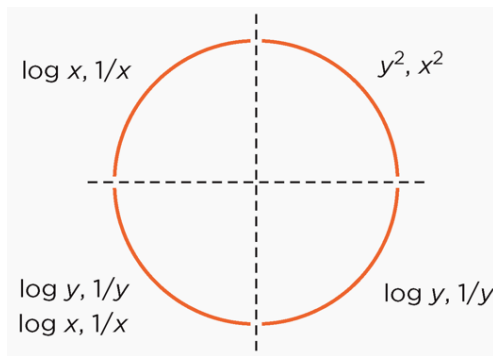
Transformations

- ▶ Transformation: re-expression of a variable by applying a function to each observation.
- ▶ Transformations allow the use of regression analysis to describe a curved pattern.
- ▶ Two nonlinear transformations useful in business applications: reciprocal and logarithms.

Choosing An Appropriate Transformation

- ▶ The process of choosing the right transformation is usually iterative.
- ▶ Among the possible choices, select the one that captures the curvature of the data and produces an interpretable equation.

Choosing An Appropriate Transformation



Match the pattern in a scatterplot of Y on X to one of the shapes to find an appropriate transformation.

└ Curved Patterns

└ Reciprocal Transformation

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Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

Regression Diagnostics

Changing Variation

Leveraged Outliers

Dependent Errors and Time Series

References

Reciprocal Transformation

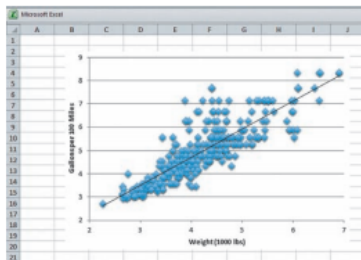
Convert Observed Data d to $1/d$

- ▶ The reciprocal transformation is useful when dealing with variables that are already in the form of a ratio, such as miles per gallon.
- ▶ In the car example, apply this transformation to the response variable and multiply by 100. The resulting response is number of gallons it takes to go 100 miles.

└ Curved Patterns

└ Reciprocal Transformation

Going back to the Example



$$\text{Estimated Gallons/Hundred Miles} = -0.11 + 1.20\text{Weight}$$

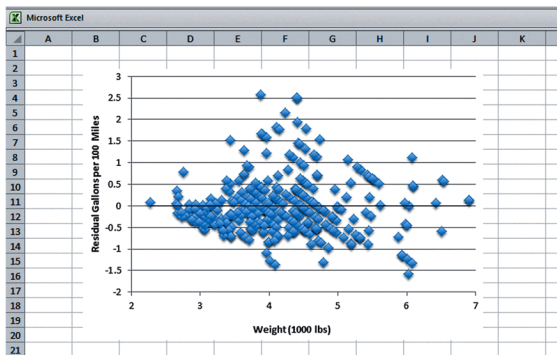
$$r^2 = 0.713; s_e = 0.667$$

The scatterplot of fuel consumption (gallons/100 miles) versus weight has a linear pattern.

- └ Curved Patterns

- └ Reciprocal Transformation

Residual Plot



Outliers apparent (i.e., exotic sports cars).

Comparing Linear and Nonlinear Equations

Which is better: the equation that regresses mileage on weight, or the equation that regresses fuel consumption on weight?

- ▶ R^2 is slightly larger for the nonlinear equation (71% to 70%); however, this is not a valid comparison since the equations have different response variables.
- ▶ Only compare R^2 between regression equations that use the same observations and same response.

Regression Diagnostics

Consider how to recognize and fix three potential problems affecting regression models: changing variation in the data, outliers, and dependence among observations

- ▶ Changing variation
- ▶ Leveraged Outliers
- ▶ Dependent errors and Time series.

Outline

Introduction

Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

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Transformations

Reciprocal Transformation

Regression Diagnostics

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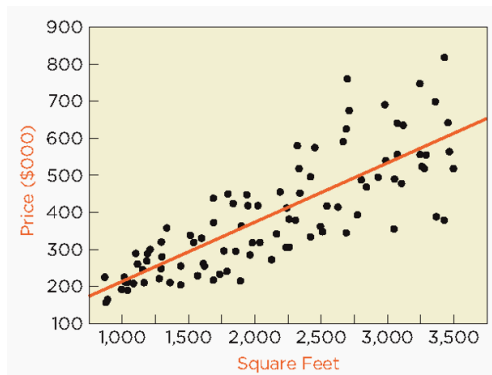
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- ▶ Although regression analysis allows the use of prices of different size homes to estimate the home of a specific size, prices tend to be more variable for larger homes. How does this affect the SRM?

Example: Price versus Home Size



- Both the average and standard deviation in price increase as home size increases.

r^2	0.6545
s_e	91.05
n	94

Term		Estimate	Std Error	t-Statistic	p-Value
Intercept	b_0	50.598687	27.41166	1.85	0.0681
Size	b_1	0.1594259	0.012076	13.20	<0.0001

- ▶ 65% of variation in prices explained
- ▶ Prediction of price of a house would be often be off by \$91,050

Observations about Intercept

- ▶ The estimated intercept (50.598687) can be interpreted as the fixed cost of a home.
- ▶ The 95% confidence interval for the intercept (after rounding) is -\$4,000 to \$105,000.
- ▶ Since it includes zero, this interval is not a precise estimate of fixed costs.

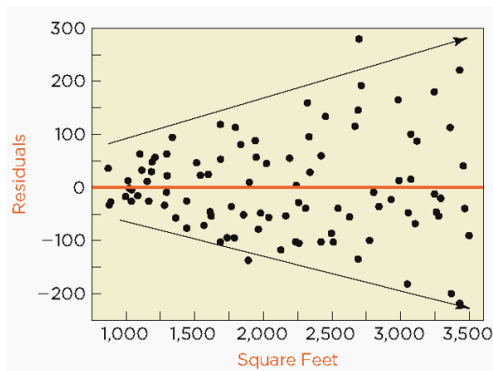
Observations about Slope

- ▶ The slope (0.1594259) estimates the marginal cost of an additional square foot of space.
- ▶ The 95% confidence interval for the slope (after rounding) is \$135,000 to \$183,500.
- ▶ It can be interpreted as the average difference in home price associated with 1,000 square feet.

Detecting Differences in Variation

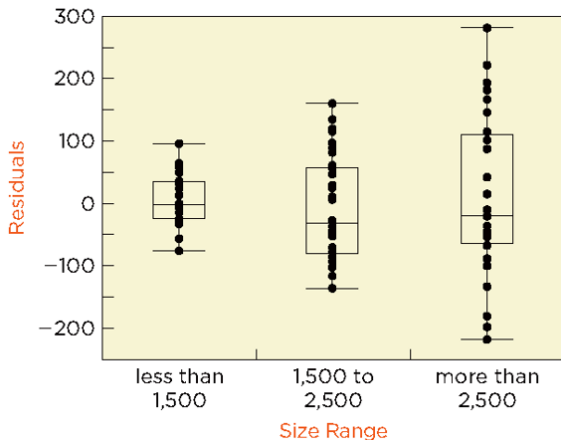
- ▶ Based on the scatterplot, the association between home price and size appears linear.
- ▶ Little concern about lurking variables since the sample of homes is from the same neighborhood.
- ▶ Similar variances condition is not satisfied.

Residual Plot



- Fan-shaped appearance of residual plot indicates changing variances.

Boxplots



- Side-by-side boxplots confirm that variances increase as home size increases.

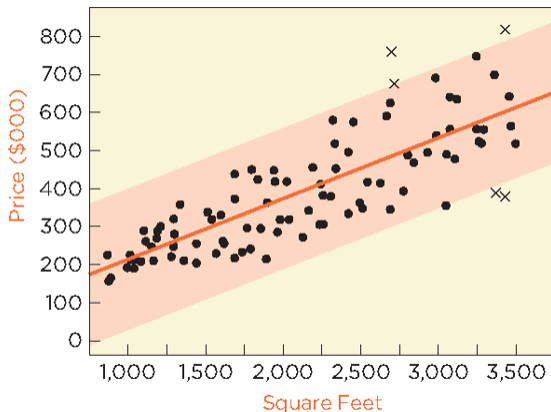
Detecting Differences in Variation

- ▶ Heteroscedastic: errors that have different amounts of variation.
- ▶ Homoscedastic: errors having equal amounts of variation.

Consequences of Different Variation

- ▶ Prediction intervals are too narrow or too wide.
- ▶ Confidence intervals for the slope and intercept are not reliable.
- ▶ Hypothesis tests regarding β_0 and β_1 are not reliable.

Consequences of Different Variation



- The 95% prediction intervals are too wide for small homes and too narrow for large homes.

Fixing the Problem: Revise the Model

- ▶ If F represents fixed cost and M marginal costs, the equation of the SRM becomes

$$\text{Price} = F + M \times \text{SqFt} + \epsilon$$

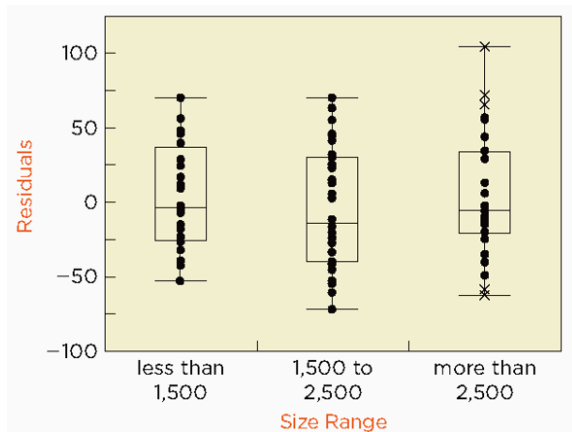
- ▶ Divide both sides of the equation by the number of square feet and simplify:

$$\frac{\text{Price}}{\text{SqFt}} = \frac{F + M \times \text{SqFt} + \epsilon}{\text{SqFt}}$$

Revised Model

- ▶ The response variable becomes price per square foot and the explanatory variable becomes the reciprocal of the number of square feet.
- ▶ The marginal cost M is the intercept and the slope is F , the fixed cost.
- ▶ The residuals have similar variances.

Boxplot



- Boxplots confirm homoscedastic errors.

Comparing Models with Different Responses

- ▶ Even though the revised model has a smaller R^2 ,
- ▶ It provides more reliable and narrower confidence intervals for fixed and variable costs; and
- ▶ It provides more sensible prediction intervals.

Comparing Models

Response	Similar Variances?	Estimated Fixed Cost	95% Confidence Interval	
			Lower	Upper
Price	No	\$50,599	−\$4,000	\$105,000
Price/Sq Ft	Yes	\$53,887	\$18,000	\$89,000

Response	Similar Variances?	Estimated Marginal Cost	95% Confidence Interval	
			Lower	Upper
Price	No	\$159/Sq Ft	\$135/Sq Ft	\$184/Sq Ft
Price/Sq Ft	Yes	\$158/Sq Ft	\$137/Sq Ft	\$179/Sq Ft

Comparing Models

Size (sq ft)	Response	Similar Variances?	95% Prediction Interval		
			Lower	Upper	Length
1,000	Price	No	\$28,000	\$392,000	\$364,000
	Price/Sq Ft	Yes	\$133,000	\$290,000	\$157,000
3,000	Price	No	\$347,000	\$711,000	\$364,000
	Price/Sq Ft	Yes	\$291,000	\$765,000	\$474,000

Outline

Introduction

Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

Regression Diagnostics

Changing Variation

Leveraged Outliers

Dependent Errors and Time Series

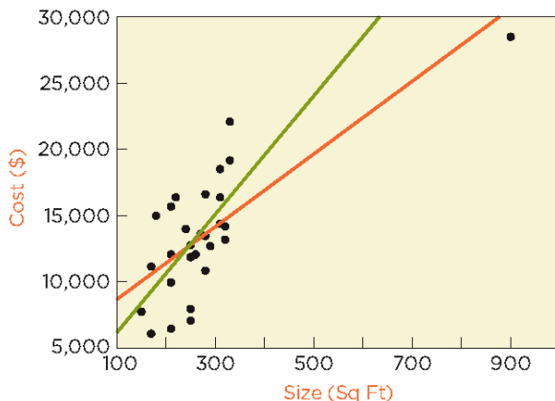
References

Consider a Contractor's Bid on a Project

A contractor is bidding on a project to construct an 875 square-foot addition to a home.

- ▶ If he bids too low, he loses money on the project.
- ▶ If he bids too high, he does not get the job.

Contractor Data for $n = 30$ Similar Projects



- Note that all but one of his previous projects are smaller than 875 square feet.

Contractor Example

- ▶ His one project at 900 square feet is an outlier.
- ▶ It is also a leveraged observation as it pulls the regression line in its direction.
- ▶ Leveraged: an observation in regression that has a small or large value of the explanatory variable

Consequences of an Outlier

- ▶ To see the consequences of an outlier, fit the least squares regression line both with and without it.
- ▶ Use the standard errors obtained without including the outlier to compare estimates.

Consequences for the Contractor Example

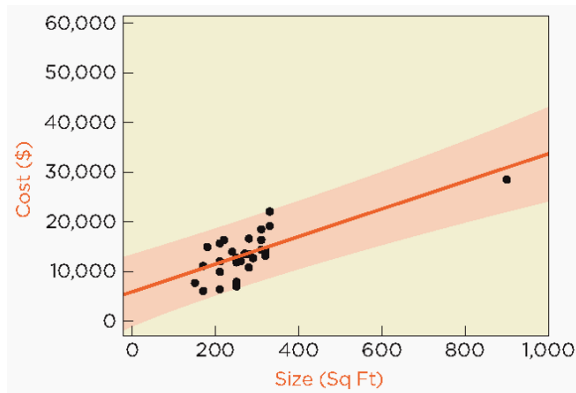
Including Outlier		Excluding Outlier
0.558567	r^2	0.376464
3,196.80	s_e	3,093.18
30	n	29

Estimate	Std Error	Term	Estimate	Std Error
5,887.74	1,400.02	b_0	1,558.17	2,877.88
27.44	4.61	b_1	44.74	11.08

Consequences for the Contractor Example

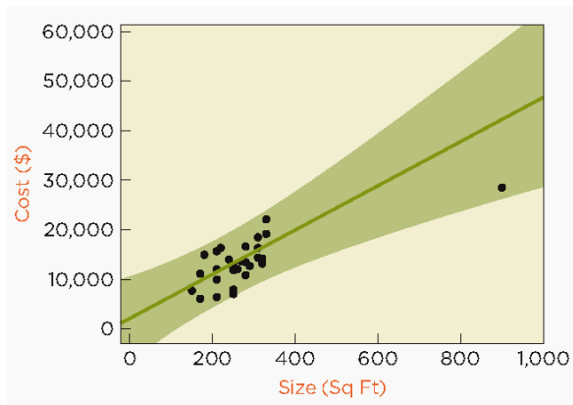
- ▶ Including the outlier shifts the estimated fixed cost up by about 1.5 standard errors.
- ▶ Including the outlier shifts the estimated marginal cost down by about 1.56 standard errors.

Consequences for the Contractor Example



- Prediction intervals when the outlier is included.

Consequences for the Contractor Example



- Prediction intervals when the outlier is not included.

Fixing the Problem

- ▶ If the outlier describes what is expected the next time under the same conditions, then it should be included.
- ▶ In the contractor example, more information is needed to decide whether to include or exclude the outlier.

Outline

Introduction

Association between Variables

Simple Regression Model

Regression for Prediction

Capital Asset Pricing Model

Inference in Regression

Prediction Intervals

Curved Patterns

Detecting Nonlinear Patterns

Transformations

Reciprocal Transformation

Regression Diagnostics

Changing Variation

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References

Detecting Dependence

- ▶ With time series data, plot residuals versus time to look for a pattern indicating dependence in the errors.
- ▶ Use the Durbin-Watson statistic to test for correlation between adjacent residuals (known as autocorrelation).

The Durbin-Watson Statistic

- ▶ Test the null hypothesis: $H_0 : \rho_t = 0$
- ▶ Is calculated as

$$D = \frac{(e_2 - e_1)^2 + (e_3 - e_2)^2 + \dots (e_n - e_{n-1})^2}{e_1^2 + e_2^2 + \dots + e_n^2}$$

- ▶ Use p -value provided by software or table to draw a conclusion.

	Reject $H_0: \rho_e = 0$ if	
n	D is less than	D is greater than
15	1.36	2.64
20	1.41	2.59
30	1.49	2.51
40	1.54	2.46
50	1.59	2.41
75	1.65	2.35
100	1.69	2.31

- Roughly, the test rejects H_0 when $D < 1.5$ or $D > 2.5$.

Consequences of Dependence

- ▶ If there is positive autocorrelation in the errors, the estimated standard errors are too small.
- ▶ The estimated slope and intercept are less precise than suggested by the output.
- ▶ Best remedy is to incorporate the dependence into the regression model.

Best Practices

- ▶ Make sure that your model makes sense.
- ▶ Plan to change your model if it does not match the data.
- ▶ Report the presence of and how you handle any outliers.

Pitfalls

- ▶ Do not rely on summary statistics like R^2 to pick the best model.
- ▶ Don't compare R^2 between regression models unless the response is the same.
- ▶ Do not check for normality until you get the right equation.
- ▶ Never forget to look at plots of the data and model.

References

1. Robert E Stine and Dean Foster, Statistics for Business: Decision Making and Analysis, Pearson.
2. Gerald Keller, Managerial Statistics, 9 Edition, Cengage Learning.

MS 6015: Multivariate Statistical Methods for Business

Regression Diagnostics

Outline

Curved Patterns

Regression Diagnostics

- Changing Variation

- Leveraged Outliers

- Dependent Errors and Time Series

Assumptions-Summary

Assumption	Diagnostic	Problem
Linearity	Scatter Plots	Nonlinear
Independence	Plots showing trend	Autocorrelation
Constant Variance	Residual Plots	Heteroscedasticity
Normality	Quantile Plots	Outliers, Skewness

Estimates of SLR Model

- ▶ $\hat{\beta}_1 = b_1 = r_{XY} \frac{s_X}{s_Y}$
- ▶ $\hat{\beta}_0 = b_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$
- ▶ $s^2 = \frac{1}{n-2} \sum_{i=1}^n e_i^2 = \frac{SSE}{n-2}$
- ▶ $s = \frac{SSE}{n-2}$ is in same units as Y is called the regression standard error.

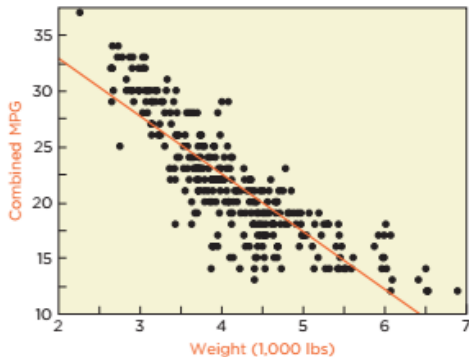
Detecting Nonlinear Patterns

- ▶ What improvement in mileage should a manufacturer expect from reducing the weight of a car?
- ▶ Use regression analysis to find an equation that summarizes the association between gas mileage and weight
- ▶ Is the pattern of association linear or curved?

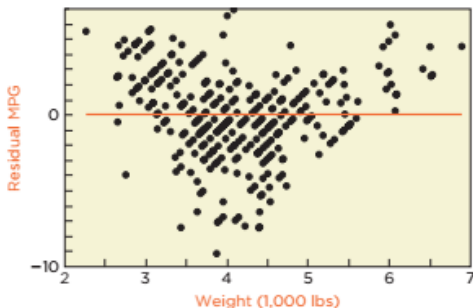
Recognizing Non linearity

- ▶ Will changes in the explanatory variable result in equal sized changes in the estimated response, regardless of the value of X ?
- ▶ Does trimming 200 pounds from a large SUV have the same effect on mileage as trimming 200 pounds from a small compact?

Scatterplot with Fitted Line



Residual Plot



Easier to spot curved pattern in the residuals. The association in the data is not straight enough to be summarized with a line.

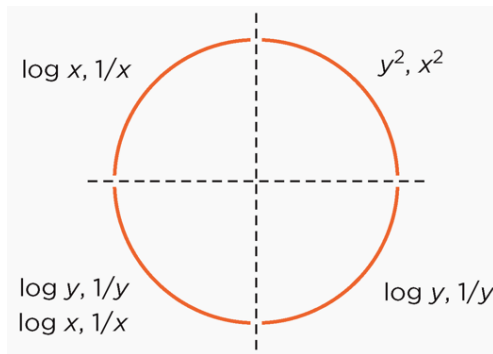
Transformations

- ▶ Transformation: re-expression of a variable by applying a function to each observation.
- ▶ Transformations allow the use of regression analysis to describe a curved pattern.
- ▶ Two nonlinear transformations useful in business applications: reciprocal and logarithms.

Choosing An Appropriate Transformation

- ▶ The process of choosing the right transformation is usually iterative.
- ▶ Among the possible choices, select the one that captures the curvature of the data and produces an interpretable equation.

Choosing An Appropriate Transformation



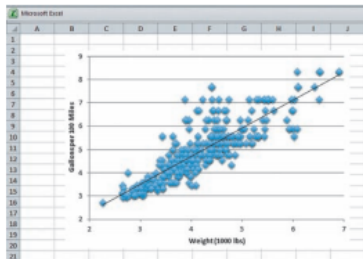
Match the pattern in a scatterplot of Y on X to one of the shapes to find an appropriate transformation.

Reciprocal Transformation

Convert Observed Data d to $1/d$

- ▶ The reciprocal transformation is useful when dealing with variables that are already in the form of a ratio, such as miles per gallon.
- ▶ In the car example, apply this transformation to the response variable and multiply by 100. The resulting response is number of gallons it takes to go 100 miles.

Going back to the Example

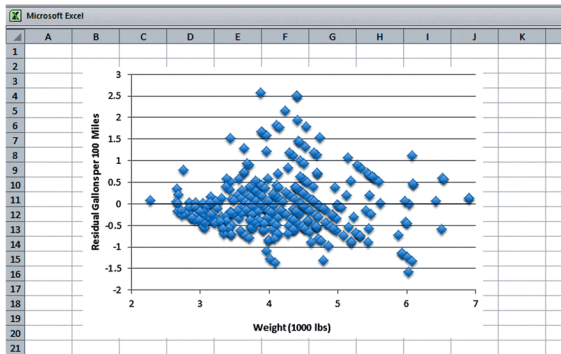


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$$r^2 = 0.713; s_e = 0.667$$

The scatterplot of fuel consumption (gallons/100 miles) versus weight has a linear pattern.

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Outliers apparent (i.e., exotic sports cars).

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Which is better: the equation that regresses mileage on weight, or the equation that regresses fuel consumption on weight?

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Consider how to recognize and fix three potential problems affecting regression models: changing variation in the data, outliers, and dependence among observations

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- ▶ Leveraged Outliers
- ▶ Dependent errors and Time series.

Outline

Curved Patterns

Regression Diagnostics

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Leveraged Outliers

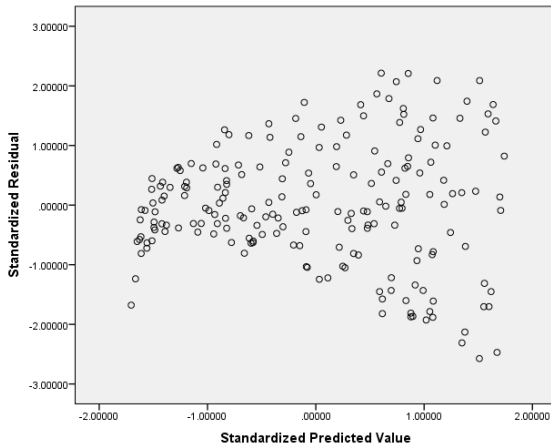
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- ▶ Based on the scatterplot, the association between home price and size appears linear.
- ▶ Little concern about lurking variables since the sample of homes is from the same neighborhood.
- ▶ Similar variances condition is not satisfied.

Residual Plot



- Fan-shaped appearance of residual plot indicates changing variances.

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Curved Patterns

Regression Diagnostics

Changing Variation

Leveraged Outliers

Dependent Errors and Time Series

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Curved Patterns

Regression Diagnostics

Changing Variation

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- ▶ Use p -value provided by software or table to draw a conclusion.

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- ▶ Best remedy is to incorporate the dependence into the regression model.

MS 6015: Multivariate Statistical Methods for Business

The Multiple Regression Model

Outline

The Multiple Regression Model

Why MLR?

- ▶ Many problems involve more than one independent variable or factor which affects the dependent or response variable.
 - ▶ More than size to predict house price.
 - ▶ Demand for a product given prices of competing brands, advertising, house hold attributes, etc.
- ▶ In SLR, the conditional mean of Y depends on X . The Multiple Linear Regression (MLR) model extends this idea to include more than one independent variable.

Multiple Regression Model

Multiple Regression Model :

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

Assumptions

- ▶ Conditional Mean of Y is linear in X_j variables.
 - ▶ Error is i.i.d with $N(0, \sigma^2)$
 - ▶ Error terms are Normally distributed.
 - ▶ Error terms have constant variance.

$$Y|X_1, \dots, X_p \sim N(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p, \sigma^2)$$

Example: Fitting a Regression Model to Real Estate data

- ▶ Model : $Y_j = \beta_0 + \beta_1 x_{j1} + \beta_2 x_{j2} + \epsilon_j$
 - ▶ X_1 = Total dwelling size(hundreds of square feet)
 - ▶ X_2 = Assessed value (in thousands of Dollars)
 - ▶ Y = Selling price(in thousands of Dollars).
- ▶ Fitted equation: $Y = 30.967 + 2.634X_1 + 0.045X_2$;
 $s = 3.473$
- ▶ Regression Standard Error
 - ▶ $s^2 = \frac{1}{n-p-1} \sum_{i=1}^n e_i^2 = \frac{SSE}{n-p-1}$
 - ▶ $\hat{\sigma} = s = \sqrt{s^2}$

- ▶ As in the SLR model, the residuals in multiple regression are on average zero.
- ▶ Because the fitted values are an exact linear combination of the X 's they are not correlated to the residuals.
- ▶ We decompose Y into the part predicted by X and the part due to idiosyncratic error; $Y = \hat{Y} + e$
- ▶ $\text{corr}(X_j, e) = 0$; $\text{corr}(\hat{Y}, e) = 0$

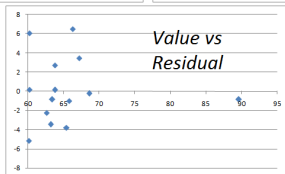
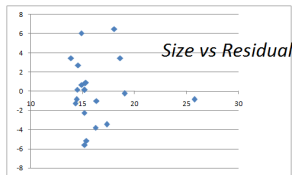
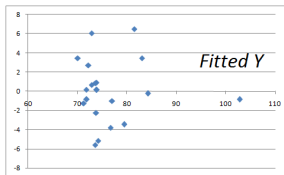
Least Square Estimates of MLR Model

- ▶ Just as before $\hat{\beta}_i$ is estimate of β_i
- ▶ $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots + \hat{\beta}_p X_p$.
- ▶ Residuals: $e_i = Y_i - \hat{Y}_i$
- ▶ Find $\hat{\beta}_i; i = 0, 1, \dots, p$ to minimize $\sum_{i=1}^n e_i^2$.

Checking Conditions

- ▶ Plot Y versus $\hat{Y}$ and verify the association appears linear.
- ▶ If pattern is linear, fit the least squares regression line and obtain residuals
- ▶ Plot residuals against $\hat{Y}$ all explanatory variables-plot should have no pattern. Note presence of outliers as well.
- ▶ If data measured over time, check for timeplot for dependence
- ▶ Inspect histogram and quantile plot of residuals to check normality.

Residual Plots



Coefficient of Determination

The coefficient of determination, denoted by R^2 measures goodness of fit:

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

tells us what is the percentage of variation in Y explained by the X 's.

- ▶ $0 < R^2 < 1$
- ▶ The closer R^2 is to 1, the better the fit.
- ▶ R^2 is the square of correlation between Y and $\hat{Y}$.

Adjusted R^2

- ▶ As we add more and more variables to the model, even random ones, R^2 will increase to 1.
- ▶ Adjusted R^2 tries to take this into account by replacing sums of squares by mean squares

$$R_a^2 = 1 - \frac{MSE}{MST}$$

Sampling Distribution

- ▶ As in SLR, the sampling distribution tells us how close we can expect $\hat{\beta}_j$ to be from β_j .
- ▶ It turns out that b_j is normally distributed (approximately):
$$b_j \sim N(\beta_j, s_{b_j}^2)$$

Confidence Intervals, t - statistic, p values

Intervals and t -statistics are exactly the same as in SLR.

- ▶ Since $b_j \sim N(\beta_j, s_{b_j}^2)$; thus
- ▶ 95% confidence interval : $b_j \pm 2 \times s_{b_j}$.
- ▶ Building a Test Statistic:

$$t = \frac{b_j - \beta_j^0}{s_{b_j}}$$

This quantity measures how many standard deviations the estimate b_j is away from the proposed value β_j^0 .

- ▶ If the absolute value of t is greater than 2, we reject the hypothesis.
- ▶ Also as before, a small p -value(< 0.05) leads to a rejection of the null

Example: Price, Sales

- ▶ Consider sales of a product as predicted by price of this product (P_1) and the price of a competing product (P_2).

$$\text{Sales} = \beta_0 + \beta_1 P_1 + \beta_2 P_2 + \epsilon$$

The Sales Data:

- ▶ Sales : units sold in excess of a baseline
- ▶ P_1 : our price in Dollars (in excess of a baseline price)
- ▶ P_2 : competitors price (again, over a baseline)
- ▶ $b_0 = \hat{\beta}_0 = 115.72$
- ▶ $b_1 = \hat{\beta}_1 = -97.66$
- ▶ $b_2 = \hat{\beta}_2 = 108.80$
- ▶ $s = \hat{\sigma} = 28.42$

Prediction??

- ▶ Suppose my competitor will charge \$10 the next quarter. After some marketing analysis I decided to charge \$8. How much will I sell?
- ▶ $Sales = 115 - 97 * 8 + 109 * 10 + \epsilon = 437 + \epsilon$
- ▶ $Sales|P_1 = 8, P_2 = 10 \sim N(437, 28^2)$
- ▶ 95% Prediction Interval : $437 \pm 2 * 28 = (381, 493)$

Understanding dependencies of X 's

- ▶ How dependencies between the X 's affect our interpretation of a multiple regression?
- ▶ If we regress Sales on our own price, we obtain a somewhat surprising conclusion... the higher the price the more we sell!!

Understanding MLR

- ▶ The regression equation for Sales on own price (P_1) is:

$$\text{Sales} = 211 + 63.7P_1$$

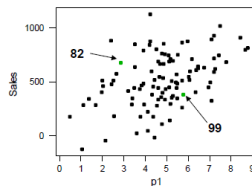
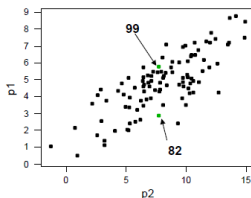
- ▶ If now we add the competitors price to the regression we get

$$\text{Sales} = 116 - 97.7P_1 + 109P_2$$

- ▶ -97.7 is the affect on sales of a change in P_1 with P_2 held fixed.

Understanding Multiple Regression

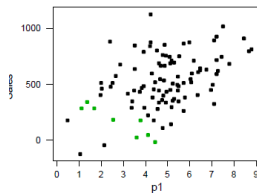
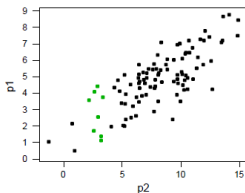
- ▶ Compare Sales in two different observations: weeks 82 and 99.
- ▶ Observe that an increase in P_1 , holding P_2 constant, corresponds to a drop in Sales.



Note that there is a strong relationship between P_1 and P_2 .

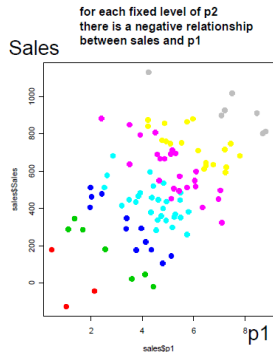
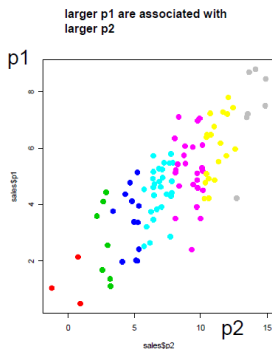
Understanding Multiple Regression

Look at a subset of points where P_1 varies and P_2 is held approximately constant.



Notice that for a fixed level of P_2 , variation in P_1 is negatively correlated with sales.

Understanding Multiple Regression



Observations

1. A larger P_1 is associated with larger P_2 and the overall effect leads to bigger sales
2. With P_2 held fixed, a larger P_1 leads to lower sales

Dependencies and Standard Error

- ▶ We saw how the relationship amongst the X 's can affect our interpretation of a multiple regression.
- ▶ We will now look at how these dependencies will inflate the standard errors for the regression coefficients, and hence our uncertainty about them.
- ▶ In Simple Linear Regression uncertainty about b_1 is measured by

$$s_{b_1}^2 = \frac{s^2}{(n-1)s_X^2} = \frac{s^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Dependencies and Standard Error

- ▶ In Multiple Regression we seek to relate the variation in Y to the variation in an X holding the other X 's fixed. So, we need to see how much each X varies on its own.
- ▶ in MLR, the standard errors are defined by the following formula:

$$s_{b_j}^2 = \frac{s^2}{\text{Variation in } X_j \text{ not associated with other } X\text{'s}}$$

- ▶ How do we measure the bottom part of the equation?

Regress X_j on all other X 's and compute the residual sum of squares(call it SSE_j) so that

$$s_{b_j}^2 = \frac{s^2}{SSE_j}$$

Multicollinearity

- ▶ If we regress Y on X 's that are highly correlated. SSE_j goes down and the standard error s_{b_j} goes up.
- ▶ Since s_{b_j} goes up, the confidence intervals $b_j \pm 2 \times s_{b_j}$ get wider.
- ▶ This situation is called "Multicollinearity".

F - tests

- ▶ In many situation, we need a testing procedure that can address simultaneous hypotheses about more than one coefficient
- ▶ Why not the t -test?
- ▶ Two important types of simultaneous tests
 - ▶ Overall Test of Significance: determine whether or not our regression is worth anything
 - ▶ Partial F-test: to compare different models.

Why not look at R^2

- ▶ To answer the question: whether or not our regression is worth anything ; Why not look at R^2 ?
- ▶ R^2 always grows as we increase the number of variables.
- ▶ Even if there is no relationship between X 's and Y , $R^2 > 0$.
- ▶ Adjusted R^2 tries to take this into account by replacing the sum of squares by mean squares

$$R_a^2 = 1 - \frac{SSE/(n - p - 1)}{SST(n - 1)} = 1 - \frac{MSE}{MST}$$

- ▶ There is no theory for inference about R_a^2 , so we will not be able to tell "how big is big".

f statistic

Define

$$f = \frac{R^2/p}{(1 - R^2)/(n - p - 1)}$$

It is also equal to

$$f = \frac{SSR/p}{SSE/(n - p - 1)} = \frac{MSR}{MSE}$$

A big f corresponds to a big R^2
"How big is big?"

Testing

- ▶ Suppose we are testing

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$$

$$H_1 : \text{at least one } \beta_j \neq 0$$

- ▶ This is the F test of overall significance
- ▶ Under the null hypothesis f is distributed as

$$f \sim F_{p, n-p-1}$$

- ▶ The p -value for the F -test is

$$p\text{-values} = \Pr(F_{p, n-p-1} > f)$$

- ▶ Reject Null if we have big f , small p -value

Generally $f > 4$ is very significant $\Rightarrow$ Reject Null !!

Inference in Multiple Regression

1. We test whether the overall model explains more variation than we would expect by chance alone.
2. If it does, then we test contributions of individual contributory variables.

F -Test

1. Inference for the Model: F -test
2. F -test: test of the explanatory power of the MRM as a whole.
3. F -statistic: ratio of the sample variance of the fitted values to the variance of the residuals.

Steps in fitting a Multiple Regression

1. What is the problem to be solved? Do these data help in solving it?
2. Check the scatterplots of the response versus each explanatory variable (scatterplot matrix).
3. If the scatterplots appear straight enough, fit the multiple regression model. Otherwise find a transformation.
4. Obtain the residuals and fitted values from the regression.

Steps in Multiple Regression- contd.

5. Use residual plot of e vs. $\hat{y}$ to check for similar variance condition.
6. Construct residual plots of e vs. individual explanatory variables. Look for patterns.
7. Check whether the residuals are nearly normal.
8. Use the F -statistic to test the null hypothesis that the collection of explanatory variables has no effect on the response.
9. If the F -statistic is statistically significant, test and interpret individual partial slopes.

Partial F -tests

- ▶ What about fitting a reduced model with only a couple of X 's? In other words, do we need all of the X 's to explain Y ?
- ▶ Can we make a decision based only in the R^2 calculations?
- ▶ With the total F -test, we were asking :Is this regression worthwhile?
- ▶ Now, we're asking :Is is useful to add these extra covariates to the regression?
- ▶ You always want to use the simplest model possible. : Only add covariates if they are truly informative.

Partial F -test

- Consider the regression model

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_{p_{base}} X_{p_{base}} + \beta_{p_{base}+1} X_{p_{base}+1} + \dots + \beta_{p_{full}} X_{p_{full}}$$

such that p_{base} is the number of covariates in the base model and $p_{full} > p_{base}$ is the number of covariates in the full model.

- The partial F -test is concerned about the hypothesis

$$H_0 : \beta_{p_{base}+1} = \beta_{p_{base}+2} = \dots = \beta_{p_{full}} = 0$$

$$H_1 : \text{at least one of the } \beta_j \neq 0 \text{ for } j > p_{base}$$

Partial F -test

It turns out that under the Null hypothesis (i.e. base model is true)

$$f = \frac{(R_{full}^2 - R_{base}^2)/(p_{full} - p_{base})}{(1 - R_{full}^2)/(n - p_{full} - 1)} \sim F_{p_{full} - p_{base}, n - p_{full} - 1}$$

- ▶ Big f means that $R_{full}^2 - R_{base}^2$ is statistically significant.
- ▶ Big f means that at least one of the added X 's is useful.

Observations

The R^2 will always increase as more variables are added

- ▶ If you have more b 's to tune, you can get a smaller SSE .
- ▶ Least squares is content fit "noise" in the data.
- ▶ This is known as **overfitting**.

More parameters will always result in a better fit to the sample data, but will not necessarily lead to better predictions.

F Test vs t - test

- ▶ Why not always decide what to include by just looking at individual t -stats?
- ▶ Multiple Testing: If you use $\alpha = 0.05 = 1/20$ to judge significance, then you expect to reject a true null $\beta = 0$ about once every 20 tests.
- ▶ If the X 's are highly correlated with each other, then s_{b_j} 's will be very big, and you will fail to reject $\beta_j = 0$ for all of the X_j even if they do not have a strong effect on Y .

Best Practices

- ▶ Know the context of your model.
- ▶ Examine plots of the overall model and individual explanatory variables before interpreting the output.
- ▶ Distinguish marginal from partial slopes.
- ▶ Check the overall F -statistic before looking at the t -statistics.
- ▶ Learn to recognize the presence of collinearity.

Pitfalls

- ▶ Do not confuse a multiple regression with several simple regressions.
- ▶ Do not believe that you have all of the important variables.
- ▶ Do not think that you have found causal effects.
- ▶ Do not interpret an insignificant t -statistic to mean that an explanatory variable has no effect.

MS 6015: Multivariate Statistical Methods for Business

Multiple Regression — Multicollinearity

Outline

Multicollinearity

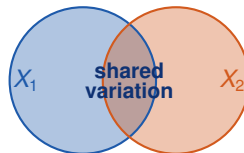
- Identifying Explanatory Variables

- Collinearity

- Removing Explanatory Variables

Introduction

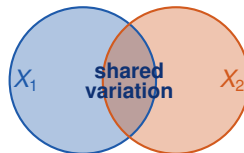
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Collinear predictors

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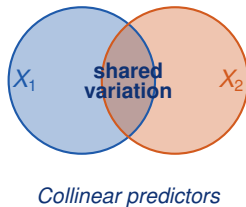
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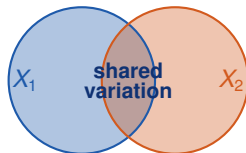
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- Regression starts with an **initial model** motivated by theory or experience.
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- **Collinearity** — correlation among predictors — **complicates this search**.



Introduction

- Regression starts with an **initial model** motivated by theory or experience.
- We seek additional variables for **better predictions**.
- **Collinearity** — correlation among predictors — **complicates this search**.
- This session: how to **detect**, **understand**, and **manage** collinearity.



Collinear predictors

- └ Multicollinearity
 - └ Identifying Explanatory Variables

Outline

Multicollinearity

Identifying Explanatory Variables

Collinearity

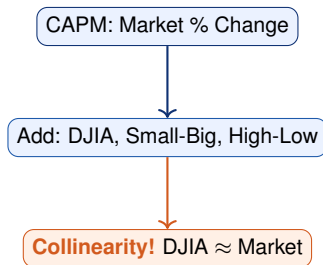
Removing Explanatory Variables

Identifying Explanatory Variables

Motivating Question

What explanatory variables belong in a regression model for stock returns?

- **Initial model** motivated by CAPM theory

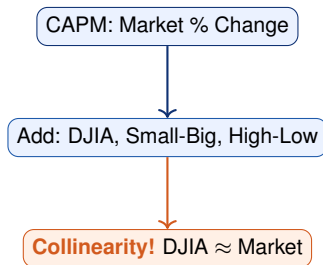


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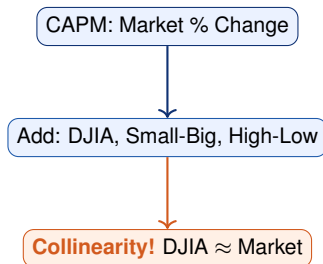


Identifying Explanatory Variables

Motivating Question

What explanatory variables belong in a regression model for stock returns?

- **Initial model** motivated by CAPM theory
- Seek variables that **improve fit** and predictions
- Process complicated by **correlated explanatory variables**



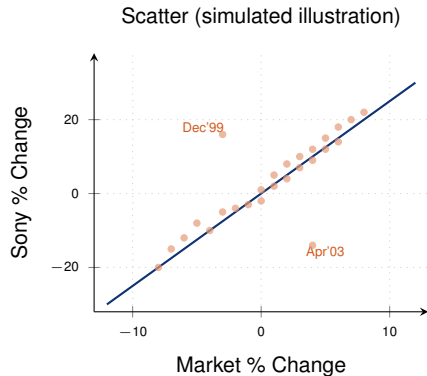
- └ Multicollinearity
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The Initial CAPM Model

Simple Regression

$$\widehat{\text{Sony}\%} = b_0 + b_1 (\text{Market}\%)$$

- Scatterplot: **linear association**, two outliers

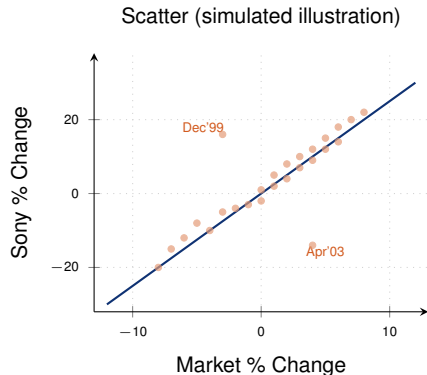


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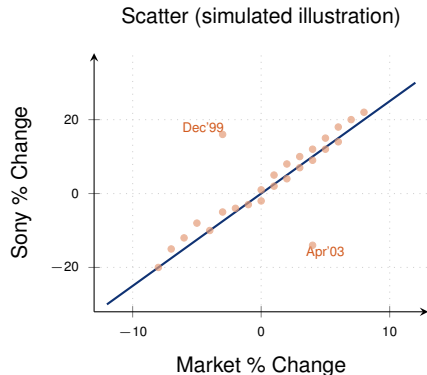


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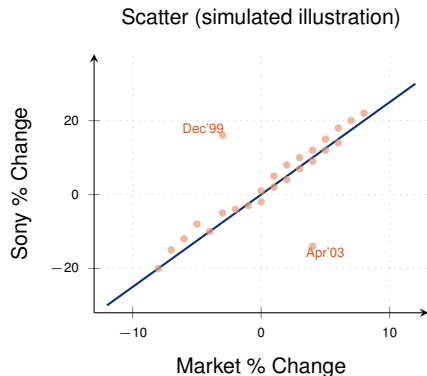
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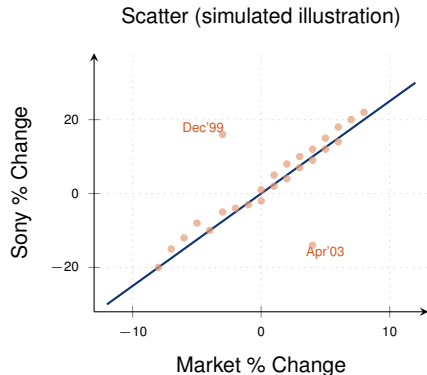
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- Normal plot: **nearly normal**
- **All conditions satisfied — proceed to inference**



Initial Model — Inference Results

Simple Regression Results ($r^2 = 0.3173$, $s_e = 8.997$, $n = 204$)

Term	$\hat{\beta}$	SE	t	p
Intercept	-0.5317	0.6338	-0.84	0.4025
Market % Change	1.2780	0.1319	9.69	<0.0001

- Intercept **not significant**
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• **Conditions met — ready to add more variables**

- └ Multicollinearity
 - └ Identifying Explanatory Variables

Correlation Among Predictors

Correlation Matrix

	Sony	Market	Dow	S-B	H-L
Sony	1.000	0.563	0.460	0.367	-0.241
Market	0.563	1.000	0.907	0.256	-0.233
Dow	0.460	0.907	1.000	0.002	-0.049
S-B	0.367	0.256	0.002	1.000	-0.362
H-L	-0.241	-0.233	-0.049	-0.362	1.000

- **Market and Dow: $r = 0.907$**
($r^2 = 82\%$ shared variation)

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- Scatterplot matrix confirms **linear associations** with response
- **All four variables included in the initial MRM**

- └ Multicollinearity
- └ Identifying Explanatory Variables

MRM with All Four Explanatory Variables

ANOVA ($R^2=0.3725$, $\bar{R}^2=0.3598$, $s_e=8.691$)

Source	df	SS	MS	F
Regression	4	8,920.49	2,230.12	29.527
Residual	199	15,030.17	75.528	
Total	203	23,950.66		

- Overall F : **highly significant**

Coefficients

Term	$\hat{\beta}$	SE	t	p
Intercept	-0.5625	0.6170	-0.91	0.3630
Market %	0.9708	0.3980	2.44	0.0156
Dow %	0.1667	0.4057	0.41	0.6816
Small-Big	0.7028	0.2090	3.36	0.0009
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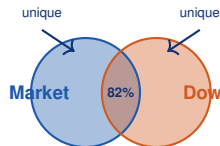
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Effect on Market % Change Estimate

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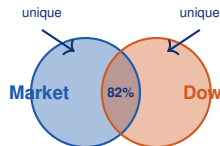
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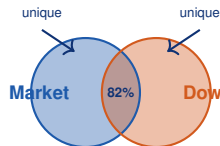
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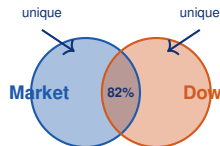
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- Market and Dow **share 82%** of variation
- Little independent signal remains to estimate separate effects
- Result: **high-SE, imprecise partial slopes**



Variance Inflation Factor (VIF)

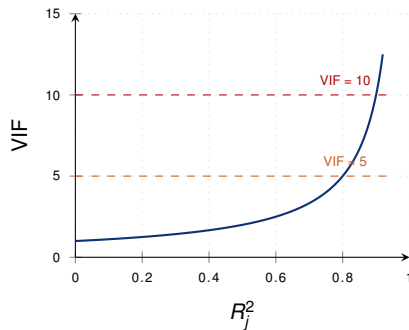
Definition

$$\text{VIF}(X_j) = \frac{1}{1 - R_j^2}$$

$R_j^2 = R^2$ from regressing X_j on all other predictors.

$$\text{se}(b_j) = \frac{s_e}{\sqrt{n} s_{X_j}} \times \underbrace{\sqrt{\text{VIF}(X_j)}}_{\text{inflation factor}}$$

VIF grows rapidly as $R_j^2 \rightarrow 1$



VIF Results — Sony Example

Full Results with VIF

Term	$\hat{\beta}$	SE	t	p	VIF
Intercept	−0.5625	0.6170	−0.91	0.3630	—
Market %	0.9708	0.3980	2.44	0.0156	9.76
Dow %	0.1667	0.4057	0.41	0.6816	8.92
Small-Big	0.7028	0.2090	3.36	0.0009	1.59
High-Low	−0.1590	0.1974	−0.81	0.4215	1.28

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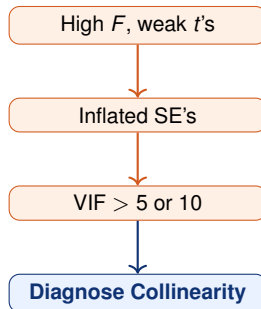
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- High-Low: VIF = 1.28 — **not redundant, just unrelated to Sony**
- Small-Big: VIF = 1.59 — **independent of other predictors**
- **Rules of thumb:**
 - VIF > 5 — moderate concern
 - VIF > 10 — serious concern

Signs of Collinearity

Warning Signs

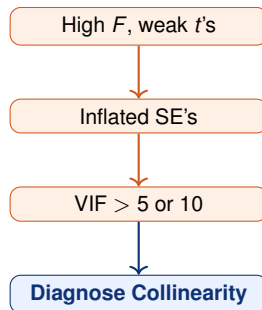
1. R^2 increases **less than expected** when adding a predictor



Signs of Collinearity

Warning Signs

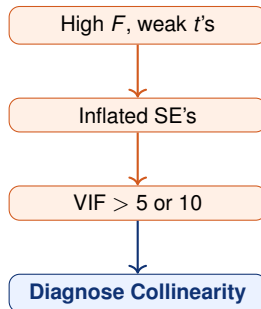
1. R^2 increases **less than expected** when adding a predictor
2. Slopes of correlated variables **change dramatically**



Signs of Collinearity

Warning Signs

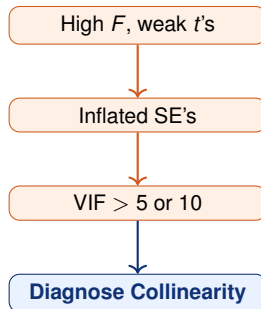
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Signs of Collinearity

Warning Signs

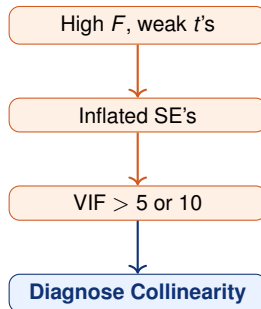
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Signs of Collinearity

Warning Signs

1. R^2 increases **less than expected** when adding a predictor
2. Slopes of correlated variables **change dramatically**
3. **F-statistic impressive** but individual t -stats are weak
4. Standard errors for partial slopes **larger than marginal SEs**
5. **Variance Inflation Factors** increase substantially



Remedies for Collinearity

Option 1 — Remove Redundant Variable

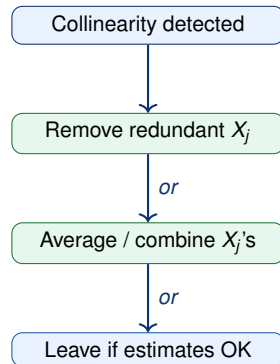
Drop Dow % Change; keep Market % (CAPM theoretical backing).

Option 2 — Re-express Variables

Use the **average** of Market % and Dow % as one predictor.

Option 3 — Do Nothing

If estimates are sensible and variables are significant, collinearity may be tolerable.



- └ Multicollinearity
 - └ Removing Explanatory Variables

Outline

Multicollinearity

Identifying Explanatory Variables

Collinearity

Removing Explanatory Variables

Issues When Removing Variables

- After adding predictors, some may become **insignificant**

Key Principle

Statistical significance alone does not determine removal.

Subject-matter knowledge must guide variable selection.

Issues When Removing Variables

- After adding predictors, some may become **insignificant**
- Remove where **statistics AND subject-matter knowledge** both agree

Key Principle

Statistical significance alone does not determine removal.

Subject-matter knowledge must guide variable selection.

Issues When Removing Variables

- After adding predictors, some may become **insignificant**
- Remove where **statistics AND subject-matter knowledge** both agree
- **Remove Dow % Change**, not Market %:
 - Market % has **CAPM theoretical grounding**
 - Dow % is largely redundant given Market %

Key Principle

Statistical significance alone does not determine removal.

Subject-matter knowledge must guide variable selection.

Best Practices and Pitfalls

- Understand the **substantive problem**
- **Plot first**: scatterplot and correlation matrices
- Use **F-test** (overall) and **t-tests** (individual)
- Learn to **recognise** collinearity signs
- **Understand** collinearity — do not fear it

- Do **not** remove variables at the **first sign** of collinearity
- Do **not** drop **several variables at once**; remove one at a time
- Do **not** ignore theory when deciding what to remove

Summary

Key Concepts

- **Collinearity**: high correlation among predictors
- Produces **inflated SEs** and **unstable slopes**
- Detected via **VIF**: > 5 caution, > 10 serious
- Warning sign: strong F , weak t 's

Take-home Message

$$\text{se}(b_j) \propto \sqrt{\text{VIF}(X_j)}$$

- Each predictor needs **unique variation**
- Theory + statistics guide **variable removal**
- Collinearity $\neq$ bad model — **manage wisely**

“Do not fear collinearity — understand it, diagnose it, and manage it deliberately.”

MS 6015: Multivariate Statistical Methods for Business

Analysis of Covariance and Regression

Outline

Two-Sample Comparisons

Analysis of Covariance

Checking Conditions

Interactions and Inference

Regression with Several Groups

Learning Objectives

To make comparisons by showing how to use regression to compare the averages of several groups.

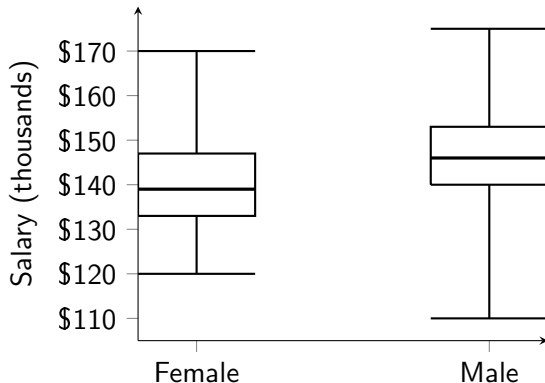
- ▶ The two-sample t -test is equivalent to a simple regression.
- ▶ A test that compares more than two groups is equivalent to a multiple regression.

Two-Sample Comparisons

Does Wal-Mart discriminate against female employees? Are they paid less than men?

- ▶ Use multiple regression with a categorical explanatory variable representing gender to analyse pay data.
- ▶ Regression analysis can adjust the comparison between men and women to account for other variables that may affect pay.

Example: Mid-Level Managers' Salaries



The average salary for women is \$140,000 and the average salary for men is \$144,700.

Example: Mid-Level Managers' Salaries

Statistic	Value
Difference $\bar{x}_m - \bar{x}_f$	4.670
Standard error $se(\bar{x}_m - \bar{x}_f)$	1.986
95% Confidence interval	0.750 to 8.590

The 95% confidence interval for the difference in mean salaries is \$750 to \$8,590 (since 0 is not in this interval, the difference is significant).

Assume conditions for inference are satisfied.

Confounding Variables

- ▶ Without a randomised experiment, we must be careful about **lurking variables** that would account for the significant difference between average salaries (e.g. experience).
- ▶ Experience is a **confounding variable** if it is correlated with salary *and* the two groups (men and women) differ with regard to experience.

Subsets and Confounding

Restrict analysis to a subset of cases with matching levels of the confounding variable (e.g. compare men and women with 5 years of experience).

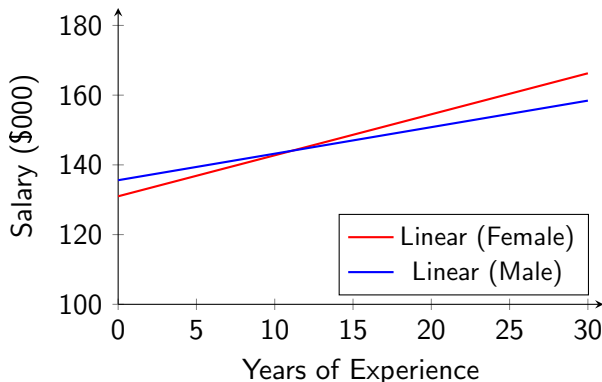
	5 Yrs Exp.	All Managers
Difference	\$378	\$4,670
Std Error of Difference	\$4,268	\$1,986
95% CI	−\$8,470 to \$9,230	\$750 to \$8,590
<i>n</i>	24	174

- ▶ The 95% CI for the 5-year subset includes 0 (not significant).
- ▶ However, the standard error is much larger; the subset does not produce a precise estimate.

Regression on Subsets

- ▶ What about the difference between average salaries for managers with 2, 10, or 15 years of experience?
- ▶ **Analysis of covariance:** regression that combines categorical and numerical explanatory variables; adjusts the comparison of means for the effects of confounding variables.

Regression on Subsets



Simple regressions fit separately show that estimated salary rises faster with experience for **women** compared to **men**.

Regression on Subsets

	Female	Male
r^2	0.139	0.093
s_e	11.225	12.059
n	59	115
Term	Estimate (SE)	Estimate (SE)
Intercept	130.989 (3.324)	135.600 (2.901)
Years	1.176 (0.388)	0.761 (0.223)

Combining Regressions

- ▶ Combining the separate regressions requires a **dummy variable** identifying gender
(Group = 1 for men; Group = 0 for women).
- ▶ Also requires the **interaction term** Group $\times$ Years.
An interaction term is the product of two explanatory variables.

Row	Sex	Group	Interaction
1	Male	1	$1 \times 16.7 = 16.7$
2	Male	1	$1 \times 6.7 = 6.7$
4	Female	0	$0 \times 13.9 = 0$
5	Female	0	$0 \times 8.5 = 0$

Combining Regressions — Output

R^2	0.135
s_e	11.785
n	174

Term	Est.	SE	t	p
Intercept	130.989	3.490	37.53	<0.0001
Years	1.176	0.408	2.89	0.0044
Group	4.611	4.497	1.03	0.3066
Group×Years	−0.415	0.462	−0.90	0.3709

Interpreting Coefficients

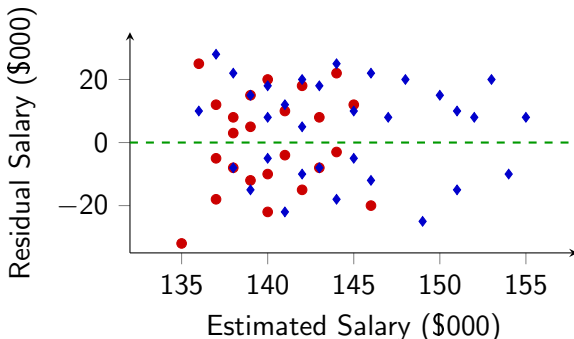
- ▶ The equation for the group coded as 0 (women) forms a **baseline** for comparison.
- ▶ The slope of the dummy variable is the difference between estimated *intercepts* in the simple regressions.
- ▶ The slope of the interaction is the difference between estimated *slopes* in the simple regressions.

Checking Conditions

- ▶ The scatterplot reveals a **linear (weak)** association between Salary and Years.
- ▶ Some caution is necessary regarding **lurking variables** (e.g. educational background or business aptitude).

Checking for Similar Variances

- ▶ Plot the residuals against the fitted values.
- ▶ Compare side-by-side boxplots of the residuals for each group. The similar variance condition is violated if the IQR in one boxplot is more than twice the length of the other.



Checking Conditions — Results

- ▶ The **similar variance condition is satisfied**.
- ▶ Examining the normal quantile plot confirms that the residuals are **nearly normal**.

Interactions and Inference

Principle of Marginality:

- ▶ If the interaction is **statistically significant**, retain it as well as both of its components regardless of their level of significance.
- ▶ If the interaction is **not statistically significant**, remove it from the regression and re-estimate the equation. A model without an interaction term is simpler to interpret since the lines fit to the groups are *parallel*.

Interactions and Collinearity

An interaction in a multiple regression introduces **collinearity** (see large VIF for Group $\times$ Years).

Term	Est.	SE	t	VIF
Intercept	130.989	3.490	37.53	—
Years	1.176	0.408	2.89	5.33
Group	4.611	4.497	1.03	5.68
Group $\times$ Years	−0.415	0.462	−0.90	13.07

Since the interaction is not significant, remove it and re-estimate the MRM.

Multiple Regression without the Interaction

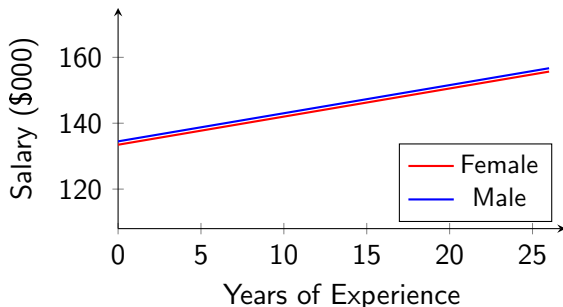
R^2	0.131
s_e	11.779
n	174

Term	Est.	SE	t	p	VIF
Intercept	133.468	2.132	62.62	<0.0001	—
Years	0.854	0.192	4.44	<0.0001	1.19
Group	1.024	2.058	0.50	0.6193	1.19

Parallel Fits

$$\mu_{\text{Salary}|\text{Years},\text{Group}} = \beta_0 + \beta_1 \text{Years} + \beta_2 \text{Group}$$

The coefficient of the dummy variable (1.024) means that the line for men is shifted up from the line for women by **\$1,024** for all levels of experience.



Parallel Fits — Conclusion

- ▶ The t -statistic and associated p -value (0.6193) for the slope of Group indicates that it is **not statistically significant**.
- ▶ This model finds **no statistically significant difference** between the average salaries of male and female managers when comparing managers with equal years of experience.

Example: Estimating Store Sales

The data measures sales of stores operated by a retail chain in three markets: **urban**, **suburban**, and **rural**.

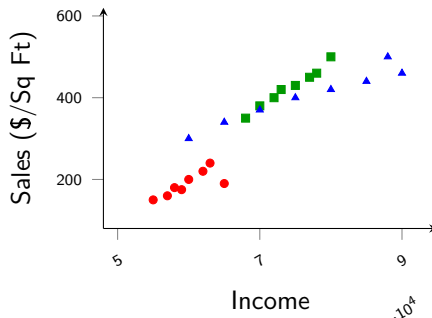
- ▶ **Explanatory variables:** median household income, local population size, market type (urban/suburban/rural).
- ▶ **Response:** sales in dollars per square foot.
- ▶ The retailer wants to compare sales per square foot across locations, accounting for substantial differences in income and population.

Analysis Steps

1. Start by plotting a scatterplot matrix — what do you observe?
2. Run regressions taking into account all numerical, categorical variables, and their interactions. Interpret the estimates.
Identify the significant factors.
3. Drop insignificant variables — re-run your regressions and interpret again.

Scatterplot Matrix

- ▶ Rural — red
- ▶ Suburban — green
- ▶ Urban — blue
- ▶ Association within each group appears **linear**.



Dummy Variables for Three Groups

In general, to distinguish J groups requires $J - 1$ dummy variables.

For this example use **two dummy variables**:

- ▶ **Suburban Dummy** = 1 if suburban, 0 otherwise
- ▶ **Urban Dummy** = 1 if urban, 0 otherwise
- ▶ Note that rural locations are coded (0, 0) — they form the baseline.

Regression Output

R^2	0.7154
s_e	62.6584
n	87

Term	Est.	SE	t	p
Intercept	−388.70	139.67	−2.78	0.0067
Income	0.0097	0.0022	4.47	<0.0001
Population	0.2401	0.0401	5.99	<0.0001
Urban Dummy	468.87	161.56	2.90	0.0048
Suburban Dummy	390.59	183.32	2.13	0.0362
Urban × Income	−0.0053	0.0025	−2.14	0.0356
Suburban × Income	−0.0068	0.0026	−2.56	0.0122

Fitted Models by Location

Coefficients associated with dummy variables reflect differences compared to **rural** stores (baseline).

Rural:

$$\hat{y} = -388.70 + 0.0097 \text{ Income} + 0.2401 \text{ Population}$$

Urban:

$$\hat{y} = 80.17 + 0.0044 \text{ Income} + 0.2401 \text{ Population}$$

Suburban:

$$\hat{y} = 1.89 + 0.0029 \text{ Income} + 0.2401 \text{ Population}$$

Interpretation of Results

- ▶ Sales at a given income are **higher in urban** compared to rural stores, but do not grow as fast with increases in income.
- ▶ **Population** has the same effect in every location because the model does not include an interaction term between Population and the location dummy variables.

MS 6015: Multivariate Statistical Methods for Business

Model Selection & Regularization

Outline

Real-World Case: Google Flu Trends

The Bias–Variance Trade-off

Model Selection

Subset Selection

- Best Subset Selection

- Stepwise Regression

Model Selection Criteria

Regularization

Summary

What is Overfitting?

Definition

A model **overfits** when it learns the training data *too well* — capturing noise and idiosyncrasies rather than the underlying pattern — and consequently performs poorly on unseen (test) data.

Signs of overfitting

- ▶ Training accuracy $\gg$ Test accuracy
- ▶ Model is overly complex (many parameters)
- ▶ Predictions are erratic on new inputs

Key metrics

- ▶ R^2 : how much variance is explained
- ▶ RMSE: average prediction error
- ▶ Gap between train & test error = *generalization gap*

Google Flu Trends (GFT) — Background

- ▶ Launched by Google in 2008 to predict flu outbreaks in real time.
- ▶ Tracked the frequency of flu-related search queries (e.g. “fever”, “cough”, “flu symptoms”).
- ▶ Out of 50 million candidate search terms, 45 terms were selected that best fitted historical CDC flu data.
- ▶ Original paper reported ~97% accuracy against historical flu seasons.

Original reference

Ginsberg et al. (2009). *Detecting influenza epidemics using search engine query data*. *Nature*, 457, 1012–1014.

GFT — The Overfitting Problem

What went wrong

The model learned *winter-season correlations*, not the biology of flu transmission. Searches for **high-school basketball** were among the top predictors — a winter activity, not a flu symptom.

- ▶ GFT was *part flu detector, part winter detector*.
- ▶ The 45 terms were chosen to maximise fit on past data — a classic case of overfitting to the training set.
- ▶ No out-of-sample validation was performed before deployment.

GFT — Performance on “Test Data” (the Real World)

	Training (2003–2007)	Test (2009–2013)
Accuracy vs CDC	~97%	Overestimated in 100/108 weeks
Peak flu estimate	Close to CDC	~2× CDC figure
2009 H1N1 pandemic	N/A (spring outbreak)	Completely missed

Table: Training vs. real-world performance of Google Flu Trends

- ▶ GFT predicted 11% of the US had flu at peak vs. CDC's 6%.
- ▶ It over-predicted flu in 100 out of 108 weeks between August 2011 and September 2013.
- ▶ The 2009 H1N1 struck in *spring* — outside the winter pattern the model had memorised — and GFT failed entirely.

Lessons Learned

1. Always evaluate on held-out test data before deployment.
2. Model complexity must be justified by the amount and quality of training data.
3. Spurious correlations are dangerous — high training accuracy does not imply causation or generalisability.
4. Domain knowledge matters — a statistician familiar with flu epidemiology would have flagged “basketball searches” as a nonsense predictor.
5. Big data $\neq$ better models — more data can amplify overfitting if the modelling pipeline is not carefully controlled.

Bias and Variance — Definitions

Bias

Error from **wrong assumptions** in the model. A high-bias model is too simple — it *underfits*, missing the true relationship in the data.

Example: fitting a straight line to data that is genuinely curved.

Variance

Error from **sensitivity to small fluctuations** in the training set. A high-variance model is too complex — it *overfits*, memorising noise instead of signal.

Example: the degree-9 polynomial that passes through every training point.

The trade-off

Reducing bias typically *increases* variance, and vice versa. The goal is to find a model complexity that **minimises total prediction error** on unseen data.

Decomposing Total Error

For any model, the expected prediction error on a new point can be decomposed as:

$$\underbrace{\mathbb{E}[(y - \hat{f})^2]}_{\text{Total Error}} = \underbrace{\text{Bias}^2(\hat{f})}_{\text{systematic error}} + \underbrace{\text{Var}(\hat{f})}_{\text{estimation error}} + \underbrace{\sigma_\varepsilon^2}_{\text{irreducible noise}}$$

- ▶ Bias^2 : how far the average prediction is from the truth.
- ▶ Var : how much predictions fluctuate across different training sets.
- ▶ σ_ε^2 : noise inherent in the data — **cannot be reduced** by any model.

Implication

Even a perfect model cannot drive total error below σ_ε^2 . Over-engineering a model reduces bias but inflates variance, *increasing* total error.

Managing the Trade-off — Practical Strategies

Reduce variance (fight overfitting)

- ▶ Collect more training data
- ▶ Regularisation (Ridge, Lasso, Dropout)
- ▶ Cross-validation for model selection
- ▶ Ensemble methods (bagging, random forests)
- ▶ Early stopping in iterative training

Reduce bias (fight underfitting)

- ▶ Use a more expressive model family
- ▶ Add relevant features / interactions
- ▶ Reduce regularisation strength
- ▶ Increase model depth (neural networks)
- ▶ Boosting (sequentially correct errors)

The golden rule

Choose model complexity using **held-out validation data**, not training error. Training error is always an optimistic estimate of generalisation error.

Why Do We Need Model Selection?

The standard linear model is:

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \epsilon$$

- ▶ Simple, interpretable, and often competitive in predictive performance.
- ▶ But with many predictors, **plain least squares breaks down**.

★ Key Insight

More predictors $\neq$ better model. Unchecked complexity leads to **overfitting**: excellent training fit but poor generalisation.

Three Classes of Methods

Subset Selection

Identify a *subset* of predictors related to the response; fit OLS on that reduced set.

Shrinkage

Fit all p predictors but *shrink* coefficients toward zero. Reduces variance; can select variables.

Dimension Reduction

Project p predictors into $M < p$ combinations; use those M projections as inputs.

★ Key Insight

All three approaches address the **bias–variance trade-off**: less flexibility increases bias but reduces variance, often yielding better test-set performance.

Subset Selection Overview

- ▶ Goal: find the *best* subset of the p available predictors.
- ▶ Two main strategies:
 - ▶ **Best Subset Selection** — exhaustive search.
 - ▶ **Stepwise Regression** — greedy search (forward / backward).

★ Key Insight

Subset selection keeps the *model form* identical to OLS — only the set of active predictors changes. Coefficients are not shrunk; they are either kept or dropped entirely.

Best Subset Selection — Algorithm

Step 1

Let $\mathcal{M}_0$ be the null model (intercept only).

Step 2 — for $k = 1, 2, \dots, p$

1. Fit *all* $\binom{p}{k}$ models with exactly k predictors.
2. Pick the best (smallest SSE / largest R^2) $\Rightarrow \mathcal{M}_k$.

Step 3

Choose among $\mathcal{M}_0, \mathcal{M}_1, \dots, \mathcal{M}_p$ using cross-validation, C_p , AIC, BIC, or Adjusted R^2 .

★ Key Insight

Step 2 minimises *training* error. Step 3 is critical: it estimates *test* error to protect against overfitting.

Example

For the sake of example, suppose we have $k = 4$ candidate predictors- x_1, x_2, x_3 , and x_4 -for our final regression model. Then, there are $2^4=8$ possible regression models we can consider:

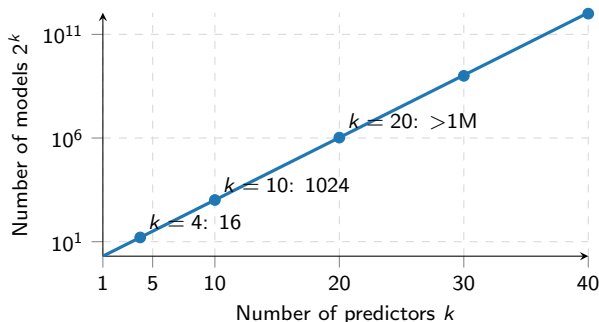
- ▶ the one (1) model with no predictors
- ▶ the three (4) models with only one predictor each.
- ▶ the three (6) models with two predictors each.
- ▶ the one (1) model with all four predictors- x_1, x_2, x_3 and x_4 .

That is $1 + 4 + 6 + 4 + 1 = 16$ possible models to consider.

In general, if there are k possible candidate predictors, then there are 2^k possible regression models containing the predictors.

Best Subset — Combinatorial Explosion

With k candidate predictors there are 2^k possible models.



△ Watch Out

Beyond $p \approx 40$, exhaustive search becomes computationally infeasible. Larger search spaces also inflate the risk of selecting spurious models.

Best Subset — Result Example

Best Subsets using Adjusted R^2 (checkmark = predictor included):

Effect	1	2	3	4	5	6	7	8	9	10
Adj. R^2	.976	.976	.975	.974	.974	.967	.964	.922	.816	.645
x1_t	✓	✓	✓	✓	✓	✓				
x2_t	✓	✓		✓	✓		✓		✓	
x4_t	✓		✓		✓	✓	✓	✓		✓
x3_t		✓	✓		✓		✓	✓	✓	

★ Key Insight

Model 1 (3 predictors: x1, x2, x4) achieves Adj. $R^2 = .976$ — almost identical to Model 2.

Stepwise regression

- ▶ Forward (selection) stepwise regression:
 - ▶ Forward stepwise selection begins with a model containing no predictors, and then adds predictors to the model, one-at-a-time, until all of the predictors are in the model. Add variable at each step based on partial F -test.
 - ▶ In particular, at each step the variable that gives the greatest additional improvement to the fit is added to the model.
- ▶ Backward (elimination) stepwise regression:
 - ▶ Like forward stepwise selection, backward stepwise selection provides an efficient alternative to best subset selection.
 - ▶ However, unlike forward stepwise selection, it begins with the full least squares model containing all p predictors, and then iteratively removes the least useful predictor, one-at-a-time. Remove variable at each step based on partial F -test.
- ▶ Stepwise regression is a combination of the forward selection and backward elimination procedures.

Stepwise Regression

Forward Selection

1. Start: null model $\mathcal{M}_0$.
2. At each step, add the predictor giving the greatest improvement.
3. Stop when no further gain.

Considers only $1 + \frac{p(p+1)}{2}$ models.

Backward Elimination

1. Start: full model $\mathcal{M}_p$.
2. At each step, remove the least useful predictor.
3. Stop when all remaining are significant.

Requires $n > p$ (full model must be fit).

★ Key Insight

Stepwise methods are **greedy** — they do not guarantee finding the globally best subset, but they scale to large p where exhaustive search is impossible.

Example: CTRP data

C_p , AIC, BIC

- ▶ While developing an MLR model, it is important to identify the ideal number of feature variables in the model. One common issue with regression models is that of **overfitting**.
- ▶ In the presence of overfitting, the model may perform well in the training data but will perform badly in the test data.
- ▶ A number of techniques for adjusting the training error for the model size are available. These approaches can be used to select among a set of models with different numbers of variables.
 1. Mallow's C_p
 2. Akaike information criterion (AIC),
 3. Bayesian information criterion (BIC)

Mallow's C_p

For a fitted least squares model containing p predictors, the C_p estimate of test MSE is computed using the equation

$$C_p = \frac{SSE_p}{MSE_{full}} - (n - 2p)$$

where SSE_p is the sum of squared errors with p parameters in the model (including constant), MSE_{full} is the mean squared error with all variables in the model and n is the number of observations.

- ▶ The best regression model is the model with a number of parameters closest to the C_p value.

Akaike information criterion (AIC)

- ▶ The AIC criterion is defined for a large class of models fit by maximum likelihood

$$AIC = -2\log L + 2.p$$

where L is the maximized value of the likelihood function for the estimated model.

- ▶ In the case of the model with Gaussian errors, maximum likelihood and least squares are the same thing. In this case AIC is given by

$$AIC = \frac{1}{n\hat{\sigma}^2}(SSE + 2p\hat{\sigma}^2)$$

where $\hat{\sigma}^2$ is an estimate of the variance of error ϵ associated with the response measurement.

Bayesian information criterion (BIC)

- ▶ BIC is derived from a Bayesian point of view, but ends up looking similar to C_p (and AIC) as well. For the least squares model with p predictors, the BIC is given by

$$BIC = \frac{1}{n}(SSE + \log(n)p\hat{\sigma}^2)$$

- ▶ Since $n > 2$ for any $n > 7$, the BIC statistic generally places a heavier penalty on models with many variables, and hence results in the selection of smaller models than C_p .

Adjusted R^2

The adjusted R^2 statistic is another popular approach for selecting among a set of models that contain different numbers of variables.

$$\text{Adjusted } R^2 = 1 - \frac{SSE/(n - p - 1)}{TSS/(n - 1)}$$

A large value of adjusted R^2 indicates a model with a small test error.

- ▶ Unlike the R^2 statistic, the adjusted R^2 statistic pays a price for the inclusion of unnecessary variables in the model.

Validation and Cross-Validation

- ▶ We can directly estimate the test error using the validation set and cross-validation methods
- ▶ Compute the validation set error or the cross-validation error for each model under consideration, and then select the model for which the resulting estimated test error is smallest.
- ▶ This procedure has an advantage relative to AIC , BIC , C_p , and adjusted R^2 , in that it provides a direct estimate of the test error, and makes fewer assumptions about the true underlying model.
- ▶ Cross-validation is a very attractive approach for selecting from among a number of models under consideration.

Model Selection Criteria

Mallow's C_p

$$C_p = \frac{\text{SSE}_p}{\text{MSE}_{\text{full}}} - (N - 2p)$$

Best model: p closest to C_p .

AIC (Gaussian errors):

$$\text{AIC} = \frac{1}{n\hat{\sigma}^2}(\text{SSE} + 2p\hat{\sigma}^2)$$

Lower is better.

BIC (heavier penalty):

$$\text{BIC} = \frac{1}{n}(\text{SSE} + \log(n) p\hat{\sigma}^2)$$

Prefers *smaller* models than AIC when $n > 7$.

Adjusted R^2 :

$$\bar{R}^2 = 1 - \frac{\text{SSE}/(n - p - 1)}{\text{SST}/(n - 1)}$$

Penalises unnecessary variables.

★ Key Insight

BIC's $\log(n)$ penalty grows with sample size, making it asymptotically **consistent** (selects the true model). AIC targets predictive accuracy. They can disagree — report both.

Subset Selection — Summary

Method	Models checked	Optimal?
Best Subset	2^p	Guaranteed
Forward	$1 + p(p + 1)/2$	Not guaranteed
Backward	$1 + p(p + 1)/2$	Not guaranteed

Criteria for final selection: C_p , AIC, BIC, Adj. R^2 , Cross-Validation.

★ Key Insight

All subset methods keep coefficients at their OLS values for the selected variables. They make a **hard** selection (in or out). Regularization methods make a **soft** selection (shrink, possibly to zero).

Introduction

- ▶ The subset selection methods use least squares to fit a linear model that contains a subset of the predictors.
- ▶ As an alternative, we can fit a model containing all p predictors using a technique that constrains or regularizes the coefficient estimates, or equivalently, that shrinks the coefficient estimates towards zero.
- ▶ It may not be immediately obvious why such a constraint should improve the fit, but it turns out that shrinking the coefficient estimates can significantly reduce their variance
- ▶ The two best-known techniques for shrinking the regression coefficients towards zero are ridge regression and the lasso.

Why Regularization?

Datasets are often:

- ▶ **High-dimensional**: p can exceed n (sensors, genomics, spectra).
- ▶ **Noisy**: measurement error inflates variance.
- ▶ **Collinear**: correlated process variables destabilise OLS.

Recall OLS: $\hat{\beta} = (X^T X)^{-1} X^T Y$. When $p > n$ or $X^T X$ is nearly singular, $(X^T X)^{-1}$ blows up $\Rightarrow$ huge, unstable coefficients.

△ Watch Out

High R^2 + large SE + low p -values is a red flag for OLS overfitting due to multicollinearity. Never interpret OLS coefficients in isolation.

Lp Penalty Norms

For regularization with the L_p norm:

$$L_p = \left(\sum_i \|\beta_i\|^p \right)^{1/p}$$

L1 (Lasso):

$$L_1 = \sum_{k=1}^p |\beta_k|$$

Diamond-shaped constraint region.

Corners at axes $\Rightarrow$ exact zeros.

L2 (Ridge):

$$L_2 = \sqrt{\sum_{k=1}^p \beta_k^2}$$

Spherical constraint region.

Coefficients approach but rarely reach zero.

★ Key Insight

The geometry of the L1 constraint diamond causes the optimal solution to land on a **corner** (sparse solution), while the L2 sphere never touches an axis unless $\lambda \rightarrow \infty$.

Ridge Regression

Minimise the **penalised** sum of squares:

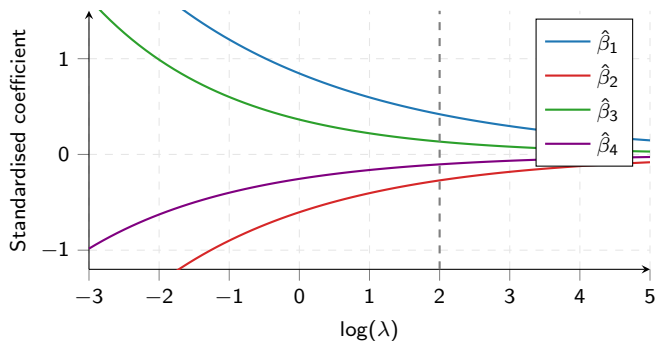
$$(Y - X\beta)^T(Y - X\beta) + \lambda \beta^T \beta \longrightarrow \hat{\beta}_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T Y$$

- ▶ Adding λI makes the matrix invertible even when $p \geq n$.
- ▶ $\lambda = 0$: recovers OLS. $\lambda \rightarrow \infty$: all $\hat{\beta} \rightarrow 0$.

★ Key Insight

The λI term is sometimes called *ridge* because it adds a “ridge” to the diagonal of $X^T X$, stabilising the inversion. This is why ridge regression works even with perfectly collinear predictors.

Ridge — Coefficient Paths



△ Watch Out

Ridge **never sets a coefficient to exactly zero** — all p predictors remain in the model. It is a *shrinkage* method, not a *selection* method. Always standardise predictors before fitting ridge.

Lasso Regression

Replace the L2 penalty with an L1 penalty:

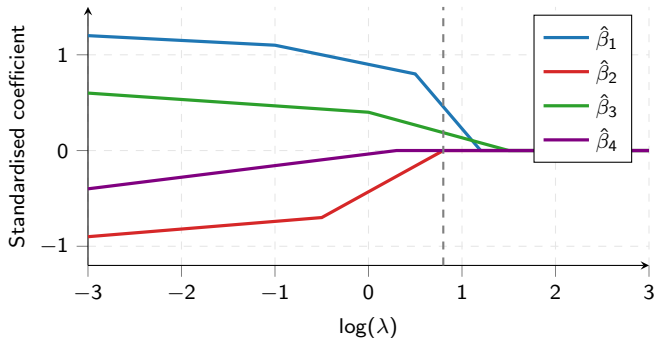
$$(Y - X\beta)^T(Y - X\beta) + \lambda|\beta|_1, \quad |\beta|_1 = \sum_{j=1}^p |\beta_j|$$

- ▶ Same tuning parameter λ controls sparsity.
- ▶ No closed-form solution (absolute value is non-differentiable at 0).
- ▶ Solved via coordinate descent or LARS algorithm.

★ Key Insight

The L1 penalty acts as a **soft-threshold**: small coefficients are driven to *exactly* zero, performing automatic variable selection. This makes the Lasso ideal for high-dimensional data where many predictors are irrelevant.

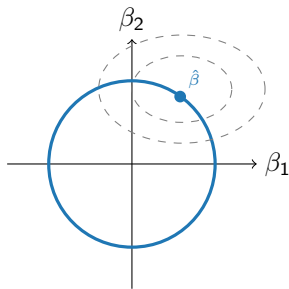
Lasso — Coefficient Paths



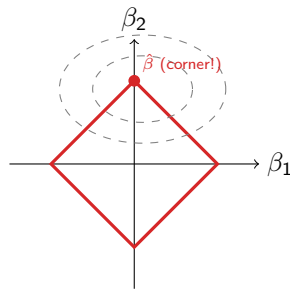
★ Key Insight

Lasso paths are **piecewise linear**: coefficients are exactly zero for large λ and “turn on” one by one as λ decreases. The order of entry reveals predictor importance.

Ridge vs. Lasso — Geometric Intuition



Ridge (L2 ball)



Lasso (L1 diamond)

★ Key Insight

The ellipses are RSS contours. Ridge's solution lies on the smooth sphere (never exactly on an axis). Lasso's solution hits a **corner of the diamond**, forcing $\beta_2 = 0$ — automatic feature selection.

Elastic Net

Combines L1 and L2 penalties:

$$(Y - X\beta)^T(Y - X\beta) + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

When to use:

- ▶ Predictors are *correlated* (groups of predictors).
- ▶ You want both *stability* (Ridge) and *sparsity* (Lasso).
- ▶ Lasso alone is unstable when $p \gg n$.

Behaviour:

- ▶ L2 part groups correlated predictors together.
- ▶ L1 part then selects among groups.
- ▶ Two hyperparameters to tune: λ_1, λ_2 .

★ Key Insight

Bond yield spread (the extra return over a risk-free rate) depends on many firm-level and macro variables — and these variables are heavily intercorrelated. Elastic Net handles this better than pure Lasso, which arbitrarily picks one from each correlated group.

Ridge vs. Lasso — Side-by-Side Comparison

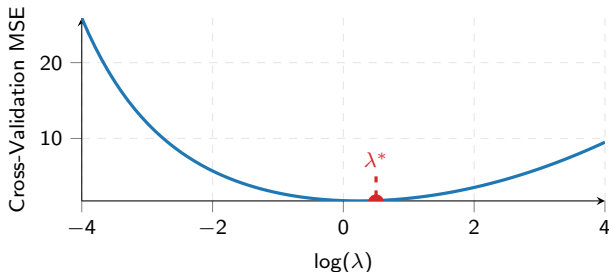
Property	Ridge (L2)	Lasso (L1)
Penalty	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $
Closed form	Yes	No (coord. descent)
Variable selection	No	Yes
Exact zeros	No ($\lambda < \infty$)	Yes
Grouped predictors	Stable	Picks one arbitrarily
Geometry	L2 sphere	L1 diamond
Best when	All predictors matter	Few predictors matter

★ Key Insight

Neither Ridge nor Lasso universally dominates. Use [cross-validation](#) to pick the best method (and λ) for your specific dataset.

Choosing λ via Cross-Validation

1. Define a grid of λ values (e.g. 10^{-4} to 10^4).
2. For each λ , compute k -fold cross-validation MSE.
3. Select $\lambda^* = \arg \min_{\lambda} CV(\lambda)$.
4. Re-fit the model on *all* data using λ^* .



★ Key Insight

The “one standard-error rule”: choose the *largest* λ whose CV error is within one SE of the minimum. This produces a sparser, more interpretable model with comparable performance.

Overall Summary

Method	Mechanism	Use when
Best Subset	Exhaustive OLS search	Small p (≤ 30)
Forward/Backward	Greedy OLS search	Moderate p
Ridge	L2 shrinkage, no zeros	All predictors relevant; collinearity present
Lasso	L1 shrinkage, exact zeros	Sparse true model; need interpretability
Elastic Net	L1 + L2 combined	Correlated predictors, $p \gg n$

★ Key Insight

Golden rule: Always use cross-validation to tune λ and to compare methods. The best technique depends on the data—there is no universal winner.

Key Takeaways

1. **Subset selection** makes hard (in/out) decisions; best for small p .
2. **Regularization** makes soft decisions; scales to large p .
3. **Ridge** stabilises correlated predictors but retains all variables.
4. **Lasso** performs automatic variable selection via exact zeros.
5. **Elastic Net** combines both for correlated high-dimensional data.
6. **Cross-validation** is essential for tuning λ and comparing methods.
7. Always **standardise** predictors before applying Ridge or Lasso.
8. The **bias–variance trade-off** underlies all regularization: a little bias can greatly reduce variance and improve test performance.